Analyzing The Wage Differential Between Formal And Informal Employment In Nepal. An Oaxaca-Blinder Decomposition Approach

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1. INTRODUCTION

1.1 Background

The informal sector of the economy is associated with lower earnings, higher risks, limited economic opportunities and legal protection as well as unable to exercise their economic rights and collective voice. International Labour Organization refers to Informal Economy as '…all economic activities by workers and economic units that are- in laws or in practice – not covered or insufficiently covered by formal arrangements, while restricting these economic activities to income generating activities involving the sale of legal goods and services (Charmes, 2012). Over 61% of people who are employed worldwide make their living in the informal economy while remaining outside the protection and regulation of the state as an informal employee (Gërxhani, 2004). The need for the understanding of the size and composition of the informal economy, what drives or causes it, and what linkages exist between formality and growth, poverty and inequality remains a major policy discourse in the study of informality. (Chen & Alter, 2012). Informality is not just a distinctive element of the informal sector but also a prevalent feature of the formal sector where it coexists alongside formal employment. Workers in the informal economy are not included and addressed in the recognition, registration, registration, and protection of labor laws and social protections of a nation, which leaves them vulnerable and powerless.

In the formal economy, workers are often able to negotiate better wages and working conditions because they are protected by labor laws and have the support of trade unions and other organizations. However, workers in the informal economy often have little bargaining power and are forced to accept lower wages and poor working conditions. An employee is entitled to social security payments, paid annual leave, paid sick leave, illness or injury benefits, and parental leave if they are registered for a permanent contract. Although informal employment does not promise these job securities, it does offer more flexibility in terms of working hours. Employers in the formal and informal sectors value occupations differently as a result of this disparity. (ben Yahmed, 2018). Therefore, it is crucial to address jobs in the formal and informal sectors via distinct labor market policies.

Along with differences in market dynamics, job opportunities, and legal protection between formal and informal workers, there are also significant differences in their earnings and risk taking. The disparity is even more pronounced in developing countries, where the majority of workers are engaged in informal employment. In these countries, workers in the informal economy often face significant challenges in terms of access to education, healthcare, and other basic services. This can lead to a lack of opportunities for upward mobility and a cycle of poverty that is difficult to break. Workers in the informal economy often have limited education and skills, which makes it difficult for them to secure formal employment and earn higher wages. In addition, the lack of labor laws and regulations in the informal economy means that employers can pay workers lower wages without facing any consequences.

Nepal has struggled with economic development compared to the rest of South Asia due to a large trade deficit and an economy heavily reliant on remittances. In 2020-2021, imports made up 36.09% of GDP and remittances contributed 22.53% (MoF, 2021). The industrial sector has decreased in its contribution to the economy and the trade deficit remains high. While gross domestic product increased from RS. 1599.22 billion in 2010-11 to RS. 2382.71 billion in 2020-2021, the past decade has seen fluctuating growth rates, reaching a peak of 8.98% in 2016-17 due to increased post-earthquake infrastructure and a low of 2.09% due to the pandemic's impact on economic activity. Approximately 28.6% of the population in Nepal lives in multidimensional poverty, with a Gini coefficient of 0.31. (CBS, 2021)

According to the Report on the Nepal Labor Force survey 2017/18 (CBS, 2019), more than 60% of the 7.1 million persons who were working were men. With 41% of all jobs, the non-agricultural informal sector was the largest contributor to overall employment. 36.5 percent of all jobs were in the official non-agricultural sector. Twenty-two percent of all jobs were in informal agriculture, which also accounted for 1.3 percent of all jobs and one percent of all jobs were in private households. The sector of non-agriculture that employed the most men was the informal non-agriculture sector (45.8%), which was followed by the formal non-agriculture sector (39%) With corresponding shares of 32.3 percent, 31.8 percent, and 32.9 percent, females were fairly evenly divided among the formal non-agriculture, informal agricultural, and informal non-agriculture sectors. Comparatively, 0.6 percent of men and little under 2 percent of women worked in private households. We choose the non-agriculture informal sector as the theoretical framework of the study as measuring the agriculture sector is challenging because the majority of the sector operates under informal mechanisms.

In Nepal, about 500,000 workers enter job market each year. 93 percent of the 15.7 million persons who were employed in 2018 were working in the informal sector. A little more than

3.3 million of the 3.8 million wage workers in 2018 had formal jobs in the private sector, 180,000 had formal jobs in the public sector, and 380,000 had formal jobs in the unofficial sector. Although agriculture still employs six out of ten workers in Nepal, it only contributes to about one-third of the country's total output. More than half of the workforce does not create enough surplus to sell on the market due to the underdevelopment and low productivity of the agriculture sector, which is primarily focused on food crops (NPC, 2020).

Nepal's economy has progressively transitioned over the past two decades from one dependent on agricultural and a bigger subsistence economy to modern industry and services, which is attributable to a higher population entering the construction, manufacturing, commerce, and transportation sectors(Ruppert Bulmer et al., 2020). Even though these jobs help people live better lives in the traditional low-productivity farm work, most of them are undocumented or temporary wage jobs that don't guarantee productivity on the supply side and come with risky working conditions, no protection from associated unforeseen risks, no written contract, no social security contributions, and no paid annual leave. In an environment driven by global economic competition, it is challenging but also right to create acceptable working conditions for the informal workers to get out of precarious circumstances in the job. However, the majority of Nepal's employment and work laws, which are essentially only focused on the official sector, do not apply to the informal sector. Due to a lack of employment prospect, legal and administrative protection and entry to open markets in Nepal, there has been a significant tendency of labor force migration overseas for employment.

Following the need to formalize informal workers and promote inclusive and employment-oriented economic growth, equitable income distribution, socioeconomic equality, and eventual increases in production and sectoral productivity, Nepal's 15th five-year plan (2018-2023) aspires to reduce the share of the informal sector in employment from 63.5 percent to 50 percent by 2023/24. Providing benefits such as pensions, healthcare and compensation makes formal employment more desirable and helps reduce incentive to work in informal sector who do not have access to the same benefits and protections. Social security system can help to level the playing field between formal and informal workers by providing benefits to workers in the informal economy.. By extending social security coverage to informal workers, governments can help to reduce the wage gap between the two sectors and promote more equitable labor market outcomes.

A targeted program has been established with the goal of bringing all workers under the social security system by gradually formalizing the informal sector and making decent employment opportunities available to the entire labor force by enhancing integrity, accountability, and competition in all economic activities. Currently, 38.5% of people in the population who are 15 years of age or older are employed, and 17% of those people are enrolled in the social security program, which accounts for 11.7% of the overall budget. Nepal government intends to make considerable progress in the areas of social and economic development, including industrialization, production and productivity, social security, and inclusion leading to economic growth via providing all working-age citizens with full and productive employment possibilities, regardless of their personal and household circumstances, employment situation, or employer (NPC, 2020c).

1.2 Statement of Problem

The majority of employment in Nepal's labor market is informal, with 59.2% of all employment in non-agricultural sectors falling under this category(CBS, 2019). Informal employment as a safety net is essential to understanding the informality of the labor market. The economic impact of informal employment on Nepal’s economy includes inefficient resource allocation, tax evasion, unfair competition and labor market distortion. Although a large number of people and households in Nepal rely on informal employment, the disadvantages outweigh the benefits for each person, each household, each business, and the nation's economy as a whole. Nepal's labor and employment laws and policies have the power to have an impact on the economy and informal jobs, however, due to Nepal's shifting political landscape, a lack of funding, political pledges, and bureaucratic failures in responding to workers' pressing needs, its implementation has been inconsistent and flawed. More than half of the population works in the unorganized sector and is not covered by social security. The government is working to enroll this group of people in the social security program. In order to achieve synchronized growth across all sectors, policymakers must pay attention to this formal-employment nexus within the current varied labor force participation. From the standpoint of formulating policies, it is crucial to evaluate the formal-informal disparities as well as the variables influencing the determinants of differences.

The research questions of the study are presented as follows:

a) Is there a wage gap differential across formal and informal employment in Nepal?

b) What are the factors to explain disparity in the wage across formal informal employment nexus in Nepal?

The proposed study is aimed to analyze the differences among the determinants of wage gap earning of the male and female workers within formal- informal employment nexus in Nepal. The research objective of the study is aimed at the following:

a) The study investigates if there is a wage differential between the formal and informal employment in Nepal and the extent of such difference.
b) The paper investigates the key factors explaining the wage differential between the two employment sectors.

1.3 Significance of the Study

A gap between the structural design of the government and the actual needs of informal workers is revealed by an examination of the literature that has already been published. As a result, the risk factors has increased, strengthening the grip of informality in the country's economy. The formal-informal wage gap is essential to understanding the informality of the labor market, particularly in developing nations where there is a big informal sector and consequently a large informal labor force. The economic impact of informal employment on developing nations' economies includes inefficient resource allocation, tax evasion, unfair competition, and labor market distortion. Although a large number of people and households in Nepal rely on informal employment as a safety net, the disadvantages outweigh the benefits for each person, each household, each business, and the nation's economy as a whole due to the unstable working conditions, lack of protection from unforeseen risks, lack of a written contract, lack of social security contributions, and lack of paid annual leaves.

The negative effects of the economic, occupational, societal, and political pressures informal workers are subjected to can have a significant impact on their ability to endure shocks. Additionally, as there are no institutional safeguards or safety nets to secure their welfare and wellness, informal workers suffer a terrible lack of security for themselves and their family. Economic risks impact informal workers disproportionately as compared to formal sector workers, making them more vulnerable to negative effects of disasters and unpredictable shocks. Lack or informal sources of savings, high indebtedness, low and fluctuating income, absence of social security or minimum wage provisions majorly affect their economic status and recovery from setbacks. Absence of employment contracts, precarious and often hazardous working conditions increases informal workers vulnerabilities to health inequalities, physical and mental stress, workplace injuries, and even death.

Every Nepali citizen has the right to employment under the country's fundamental labor and employment laws and policies, which also have the power to have an impact on the economy and informal jobs. However, due to Nepal's shifting political landscape, a lack of funding, political pledges, and bureaucratic failures in responding to workers' pressing needs, its implementation has been inconsistent and flawed. In recent years, Nepal has significantly improved the rights, facilities, compensation, safety, and security of workers in a number of Nepali sectors from a previously constrained policy structure. Although being far more inclusive and encouraging, recent changes to national acts and regulations still mainly ignore the informal sector employees, and labor laws and clauses designed to safeguard them have not yet been implemented. Protection of informal employees is a new topic that has not yet been handled in the current environment, where the rights and benefits of workers in the informal sector are only recently being acknowledged.

In signing the Sustainable Development Goals on 17 pillars, Nepal government is committed to funding and achieving full and productive employment, with decent work for all, 'Decent Work and economic growth', which contains the following target: 'By 2030, achieve full and productive employment and decent work for all women and men, including for young people and persons with disabilities, and equal pay for work of equal value'. It is associated with dependable, inclusive, and sustainable economic growth as well as the promotion of a respectable working environment for the whole working-age population of the country. This encourages the formalization and growth of micro, small, and medium-sized businesses. In 2015, the same year that the UN adopted the SDGs, Nepal's new Constitution went into force. The SDGs strongly resound with the equity-based strategy and rapid development ambitions of the constitution, which was created to institutionalize the country's broad political, social, and economic developments (NPC, 2020a).

Because of its significant consequences on inequality and poverty in Nepal, growth of informality may be dangerous for the growth of a robust business sector and a well-functioning market economy. Concerning the protection of workers against the social, political, occupational and economic vulnerability of their jobs, policy discussion is necessary. To determine the current vulnerabilities of informal workers, more research is needed in this area. These facts can support advocacy for rights-based reforms in policymaking and aid to protect the rights of informal employees. The state of the informal workforce and informal employment, which is mostly "invisible" to government and policy makers, must thus be accurately assessed in order to better understand it and to avoid the negative effects of policy responses. This report provides the economic variables or components needed to address the issues surrounding the wage gap, its causes, and policy discourses to the Government of Nepal and stakeholders.

1.4 Limitations of the Study

The study incorporates data available from Nepal Labour Force Survey (NLFS III) which was conducted in 2017/18. As the present research is carried on at 2021, with job sector hampered and by the Covid-19 crisis, there remains limitation on the availability of latest data. These limitations includes data availability and quality issues, limited scope and time constraints, difficulty of analysis, and limited generalizability. Additionally, the challenges in controlling for all relevant variables, dealing with bias in data collection, accessing appropriate research participants, and addressing ethical considerations has also been experienced.. All of these limitations could impact the validity and usefulness of the findings.

2. LITERATURE REVIEW

2.1 Overview of methods

This section reviews the relevant literature. Firstly, we analyze on the literature relating to informal employment. Next, we review literature relating to wage gap discrimination for labor force. Subsequently, we review literature relating to empirical studies and national studies.

Informal employment, also simultaneously referred to as shadow, underground, grey, hidden economy has been an elusive subject matter, as the context has been defined in different contexts with different uses.

2.2 Origin and Development of Informality

For more than fifty years, there has been a continuous growing discourse on formality and informality. The Dutch anthropologist (Boeke, 1942)created a theory of a developing economy as a "dual" economy made up of a portion that operated outside of the market economy. Arthur Lewis (1954) developed an influential two-sector model of growth in the 1950s, with one sector consisting of modern capitalist firms that maximized profit and the other sector being made up of peasant households where there were differing norms for output sharing. The dual economy was incorporated into the traditional two-sector framework of equilibrium economics in the 1970s thanks to the Harris and Todaro (1970) model in development economics. The study by Keith Hart (1973) and the International Labour Office (ILO) mission to Kenya (ILO 1972) established the importance and challenges of the dichotomy, having first introduced the term “informal sector” into the development policy debate and research studies (Guha-Khasnobis et al., 2006).

The International Labour Conference (ILC) stressed in 1991 that a complete and multidimensional approach was needed to solve the informality problem by addressing its root causes as well as its symptoms. The ILO's Bureau for Workers' Activities (ACTRAV) organized a Symposium on Trade Unions and the Informal Sector in 1999 to talk about the difficulties trade unions face there and to come up with a plan to deal with them. Even though most employees around the world labor in the informal economy, almost all of them lack proper social security protection, organization, and a voice at work, according to the Director-report General's to the Conference in 2001, which returned to the issue of informality.

In 1993, the 15th ICLS defined an informal sector enterprise as a private unincorporated enterprise that is unregistered or small in terms of the number of employed persons. An unincorporated enterprise is defined as a production unit not constituted as a separate legal entity independent of the individual (or group of individuals) who owns it, and for which no complete set of accounts is kept. Meanwhile, an enterprise is defined as unregistered if it is not registered under specific forms of national legislation (e.g. factories’ or commercial acts, tax or social security law and professional groups’ regulatory acts). An enterprise is defined as small if its number of employees is below a specific threshold (e.g. five employees) determined by national circumstances (ILO, 2018) . Here, therefore, informality was defined using the enterprise as the unit of analysis.

However, the recognition that informal jobs exist in formal and informal enterprises, and that defining informality as informal enterprise missed a considerable amount of informality, in 2003, the 17th ILCS adopted the employment relationship as the unit of analysis and defined those in informal employment as persons whose main jobs lack basic social or legal protections or employment benefits and recognized that they may be found in the formal sector, informal sector or households. Persons in informal employment were seen to include: (a) own-account workers self-employed in their own informal sector enterprises; (b) employers self-employed in their own informal sector enterprises; (c) contributing family workers; (d) members of informal producers' cooperatives; (e) employees with informal jobs

The conceptual framework of informal employment as defined by the 15th International Conference of Labour Statisticians (ILO, 2018) as "the total number of informal jobs, whether carried out in formal sector enterprises, informal sector enterprises, or households, or as the total number of persons engaged in informal jobs during a given reference period" by the International Labour Organization. In the Nepal Labour Force Survey, those in informal employment are identified and the residual is those in formal employment. Informal employment includes employers, own-account workers and contributing family workers who are employed in informal sector establishments, as well as employees and paid apprentices / interns who do not have paid annual leave or sick leave benefits and whose employers do not contribute to their social security.

In a study by (Williams & Horodnic, 2019), informality across 112 countries has been explained as resulting from either economic under-development and a lack of modernisation of governance (modernisation theory), state over-interference in the free market (neo-liberal theory) or inadequate state intervention to protect workers from poverty (political economy theory). This has been explained subsequently herewith.

2.2.1 Modernisation hypothesis

It was historically believed that the existence of informal employment in poorer countries was a result of their economic underdevelopment and insufficient modernization in governance, but that it would inevitably fade away as these countries underwent economic development and modernization(Gilbert, 1998). Informal employment was believed to be a residue of premodern production during the majority of the 20th century, and it was predicted to disappear. Modern formal economies were denoted as "advanced" and "developed," whilst the informal economies were denoted as "backward" and "underdeveloped." As a result, informal employment came to be seen as a sign of underdevelopment and a phenomenon that would pass away with modernization of the political system and economic growth (Williams & Horodnic, 2019).

2.2.2 Neo-liberal hypothesis

The neo-liberal hypothesis states that economies with higher tax rates and greater levels of government intervention in the free market will have a higher prevalence of informal employment. Even though economic development and modernization were acknowledged, the prevalence of informal employment in some economies was seen as the result of excessive government interference in the economy and welfare, and participation was seen as a rational economic choice made voluntarily by employees and businesses to leave the formal economy because doing business in the latter required too much money, time, and effort ((Becker, 2010)).

2.2.3 Political economy hypothesis

The political economy theory states that in economies with fewer levels of governmental action to protect workers from employment-related discrimination, informal employment will predominate(Fernández-Kelly & Shefner, 2006). holds that, contrary to modernization theory, informality is an integral part of the processes of modernization and economic development, operating in an unregulated environment involving low-paid, insecure work performed under "sweatshop-like" conditions as a survival strategy by marginalized populations.

2.3 Labor market discrimination

Economists and sociologists have developed theories to explain discrimination in the labour market. In this section, we will review the economic literature on this subject. In particular, we will focus on the ‘taste discrimination’ theory proposed by (Becker, 2010) and the theory of ‘statistical discrimination’ introduced by (Aigner & Cain, 1977)

2.3.1 Labor market segmentation theory

The wage gap between formal and informal work can be viewed as a supplement to the factors that human capital alone cannot account for. (Doeringer & Piore, 1975)conducted research on the American labor market, which is split into a primary and secondary sector. The primary sector is distinguished by a high salary, stability, and a better working environment in terms of safety and productivity, whereas the secondary market is devoid of the aforementioned characteristics. They also hold the opinion that the primary labor market offers workers training opportunities and a mechanism for competitiveness that the secondary market does not.

2.3.2 Taste based and statistical discrimination

(Ashenfelter & Oaxaca, 1987) defines market discrimination as the comparison of wage rate of two groups F (for Formal employment) and I (for Informal employment) as:

a) As they are actually observed
b) As they would be observed in absence of discrimination

F and I would have same wage rate in the market in the absence of discrimination if they are perfect substitutable in production. Here, difference in the wage rate between F & I would be the measure of discrimination.

If the observed wages of F & I are π_f and π_i and if no discrimination exists wage would be π_f^o and π_i^o. In such situation, proportion wage discrimination against I is

$$D = \frac{\left[\frac{\pi_f}{\pi_i} - \frac{\pi_f^o}{\pi_i^o}\right]}{\left(\pi_f^o/\pi_i^o\right)}$$

$$\cong \ln\left(\frac{\pi_f}{\pi_i}\right) - \ln\left(\frac{\pi_f^o}{\pi_i^o}\right)$$

Where D is the proportionate fall in the I to the F wage ratio, for the case of absence of discrimination. Let us assume, some characteristics X determine pay in the absence of discrimination. For group F, relation between wage π_f and these characteristics is of the form

$$\ln \pi_f = \beta_f X + \mu$$

where β_f is unknown regression coefficients & μ is disturbance coefficient. The proportionate market wage discrimination is

$$D \cong Ln\left(\frac{\pi_f}{\pi_i}\right) - \beta_f\left(X_f - X_i\right)$$

where X_f & X_i represents the characteristics of F & I respectively. Here, discrimination is measured as the difference between and the proportionate wage difference if F were paid in same way as I.

2.4 Review of Empirical Studies

A considerable number of studies have been done on informal economic activities, origin and development of informal employment and subsequent wage discrimination in the labour market. First, literature concerning the labour market position of informal employment is analysed. Next, the literature on labour market discrimination will be reviewed.

2.4.1 Labor market position of informal employment

According to traditional human capital theory, earnings and, consequently, poverty are mostly determined by education and training, which explains why these factors are important in determining policy. (Kahyalar et al., 2018) is of the idea that economists have looked at the wage difference between the formal and informal sectors in an effort to determine whether informal workers make less money than their formal counterparts. According to the classic labor theory, employees choose for the informal economy because entrance restrictions keep them out of the formal one (e.g. labour market regulations). As a result, workers accept lower-paying occupations in the informal sector. Working in the informal sector reflects the worker's own decision, which is based on the benefits and drawbacks connected with each sector, claims the competitive labor market theory(Henley et al., 2009). Under this framework, the wage gap tends to disappear.

In attempt to identify the fundamental economic and social factors that are connected to greater levels of informal work, (C. C. Williams & Horodnic, 2019) presents the varying prevalence of informal employment around the world. Bivariate regressions are used in the study to evaluate the ILO database on the prevalence of informal employment in 112 countries based on macroeconomic and social circumstances. According to the study, there is an association between higher rates of informal employment and lower levels of economic underdevelopment, public sector corruption, and state intervention to protect employees from poverty. The level of development, corruption, tax rates, the size of the government, the amount of social contributions, and the percentage of poverty are all factors that influence the share of informal employment between countries.

(Yamamoto et al., 2019a)looks into how education level affects men and women's career choices in urban and rural settings. The results demonstrates that, in comparison to men and urban women, women in rural areas have more difficulties finding steady employment even when their degree of educational attainment is controlled. Additionally, our decomposition results demonstrate that regular female workers in rural areas have a significant wage disparity as a result of the effects of gender discrimination. These findings suggest that, despite having high levels of education comparable to those of men or urban women, rural women face considerable wage discrimination and fewer opportunities to secure steady employment.

The impact of education and experience level on the wage structure in Nepal's formal, unofficial, and agricultural sectors is examined by (Adhikari, Kc, et al., 2019). By using the Nepal Labor Force Survey -II data, the Mincerion wage equation and quantile regression technique have been employed to examine such an impact. Results indicate that in all three sectors, wage returns are positively correlated with education. Return to experience, however, is associated negatively with the agricultural industry. Additionally, the impact of returning to school is greater at higher quantiles along with the wage distribution between the formal and informal sectors. In the formal sector, education has a maximum impact of 4% at the 0.90 quantile.

A human capital model that uses Heckman's two-step technique to compensate for selectivity bias estimates a considerable formal informal wage gap for Bolivia in (Monsted, 2020) study. Then, using a Mincerian specification, wages were estimated, and it was discovered that there were some discrepancies between the formal and informal labor markets. While job-specific experience did not show the typical hump shape for these people, returns to education are often higher for those employed in the formal sector. Instead, wages are linearly related to experience in the current position, suggesting that longevity is more significant than productivity.

(Kahyalar et al., 2018)investigates the wage difference between formal and informal sector in the case of Turkish labour market using OLS regression and the Oaxaca- Ransom decomposition method. The results derived from the techniques indicates a wage gap in favour of formal sector workers and that the wage gap was found to be higher at the higher end of informality.

(C. Williams & Gashi, 2022) used 8533 household interview surveys in 2017 to break out the wage gap between official and informal work in Kosovo from a gender viewpoint. The net hourly salary that employees report is the dependent variable used in this paper. Informal workers are individuals who claim to be employed without a contract and who are not disclosed to the authorities in charge of tax and pension contributions. Three significant findings may be seen by looking at the descriptive data on the net salary rates of men and women in formal and informal work. First off, people in formal employment make an hourly average net salary of €2.50 as opposed to €1.40 for those in informal employment. Second, formal employment pays higher net wages to both men and women than does informal employment. Third, while women are paid 7.1 percent more than men in informal employment (€1.50 compared to €1.40), men are paid 4% more than women in formal employment. The end result is that, when working in both the formal and informal economies is considered, women earn a net hourly pay that is 9.5% greater than that of men (€2.30 against €2.10).

The wage gap was explored at the mean using OLS regression by (R. L. Oaxaca & Ransom, 1994) in decomposition method-with and without sample selection-in the study in Turkey on the wage difference between the formal and the informal sector. The results of this technique show that there is a wage disparity in favor of those employed in the formal sector, and that the salary gap was greater when the informality definition was at its upper bound. The findings show there is a wage gap along the wage distribution that favors workers in the formal sector. The findings show there is a wage gap along the wage distribution that favors workers in the formal sector.

2.5 Review of National Studies

The wage differences between formal and informal work in Nepal have not been extensively studied. The gender wage gap in Nepal's urban and rural labor markets was studied by(Yamamoto et al., 2019a). The estimation results revealed that due to the unexplained effects, rural females experience disadvantages in both job opportunities and wage payments. Even those with greater degrees have trouble finding jobs for urban women. The Oaxaca-Blinder estimation finding implies that the impacts of discrimination vary depending on job level and the geographic areas attributed to regional variations in industrial structures. While the majority of men and women work in the agriculture sector in rural areas, most females work in agriculture area and males in informal employment.

(Parajuli, 2014) analyzes the determinants of informal employment in Nepal and significance of education attainment on wage discrimination status among formal and informal sectors excluding agriculture in Nepal. He used the informality information from NLFS II data set produced by ILO ; a national level representative household surveys regarding information on formal and informal employment information. The significance of education, age, gender, marital status, occupation & urban and rural workers on hourly wage of workers in informal sector has been observed. In the study, a simple Probit regression model is estimated using MLE method to capture and analyze probability of participating informal employment instead of formal one with stated characteristics. The studies finds out individual being in Janajati and Dalit as ethnic group are likely to participate 1.14 and 1.10 times more in informal employment in reference to Bramhan. Similarly, being a male worker is associated 1.53 times more probability of being employed informally and an individual residing in urban sector has 1.17 times higher probability to work in informal sector than a person living in rural areas.

To improve cross-country comparability, the figures are based on the most recent national labor force surveys or household surveys that follow a standard definition of informal work. The significance of the three explanations for the higher levels of informal employment—economic underdevelopment and corruption (modernization theory), excessive government involvement (neo-liberal theory), and insufficient government protection of workers from poverty (political economy theory)—is examined. According to the study, lower informal employment is linked to state action in the form of higher tax rates and social transfers to shield employees from poverty and development. In Nepal, 94.3 percent of the workforce has an informal job as their primary source of income, and 90.7 percent of those people operate in the informal economy.

By incorporating occupational and firm size effects into the Oaxaca method, (Mainali et al., 2017) examines the caste wage disparities in Nepal. The study looked at the 2003 and 2010 surveys of the Nepali labor force. Along with disparities in human capital endowments, occupational and company size influences on explaining inequality, caste wage disparity is found in the Nepalese labor market. The outcome of the discriminatory components shows that for both Matwali and Pani-Nachalne groups, access disparities persist despite variations in human capital. This shows that discriminatory practices by employers still exist in Nepal, as does a lack of networks to help individuals of low caste groups get employment in higher paying companies.

2.6 Research Gap

In Nepal, study has been done regarding the size of informal economy using Currency Demand Approach (Raut et al., 2014), significance of education and experience on wage distribution using quantile regression approach(Adhikari, Kishor, et al., 2019), gender based differences in wage distribution and employment opportunities employing Oaxaca–Blinder decomposition using NLFS II data (Yamamoto et al., 2019), determinants of informal employment and wage gap differential using probit regression model (Parajuli, 2014) but as per the knowledge of the researcher, this is the first study being incorporated on the determinants of wage gap differentials between formal and informal employment in Nepal using Oaxaca-blinder decomposition from NLFS III data.

3. METHODOLOGY

3.1 Theoretical Model

Formal sector (non-agriculture) comprises those employed in government or state-owned enterprises or international organizations/foreign embassies and those working for incorporated companies or establishments that are registered with relevant authorities. Informal sector (non-agriculture), on the other hand, comprises those employed in enterprises that are neither incorporated nor registered with authorities. Those employed in private households are regarded as in the informal sector. Those whose economic activities are in the agricultural industries are reported on separately.

To analyze whether the net hourly earnings of men and women in formal employment are higher than men and women in informal employment, firstly, descriptive results are examined on the hourly wage rates of men and women participating in formal and informal employment. Secondly, these differences are analyzed using the Blinder–Oaxaca decomposition statistical method (Blinder, 1973; R. Oaxaca, 1973). The Blinder-Oaxaca decomposition is a methodology commonly utilized in the study of differences in labour market outcomes by groups such as by gender and age. This explains the difference in the means of a dependent variable between two groups by decomposing the gap into that part which is due to differences in the mean values of the independent variable within the groups, and group differences in the effects of the independent variable. Here, it is employed to analyse the wage differential between informal and formal employment.

3.2 Data

This study usages nationally representative labor force surveys conducted in 2017 referred as Nepal Labor Force Survey- third round (NLFS -III). Central Bureau of Statistics (CBS) of Nepal administers the survey following the standard methodology and sampling design suggested by World Bank's living standard measurement survey framework. Using a multi-stage clustering sampling approach, this survey collects rich information about individual labor pertaining to various labor related indicators. Nepal Labor Force Survey (2017/18) data produced by the Central Bureau of Statistics has been used for the analysis.

Our earnings measure is daily wage. The wage earner criteria reduce the observations from 77,638 to 8743. Our decomposition analysis is restricted to individuals of 15-60 years who are in full time wage employment defined as individuals who work for 40 hours or more during the week satisfying the following criteria:

(a) work in private financial business firm, private non-financial institutions, non-profit institutions and other institutions. Those who work in government, state-owned enterprises and international organizations/foreign embassy are excluded

(b) mode of payment is cash.

The dependent variable used in this paper is the net log hourly wage reported by employees. Hourly wages is computed by dividing monthly wages by the total hours of work per month. The survey collected information on the usual hours work per week but not the number of weeks worked during a month. Therefore, the monthly hours of work is computed by multiplying the usual hours of work per week to 4. Informal employees are those reporting that they are working without a contract and are not subject to paid annual leave and social security contribution by authorities. To control for other variables that determine wage levels, three groups of explanatory variables are included in the model: personal and household characteristics, job characteristics and employer characteristics.

The following personal and household characteristics are included, given that wage levels have been previously identified as significantly associated with marital status, age and education level and the presence of children and elderly in household(Becker, 2010; Borjas, 2005):

- Marital status - a dummy variable for marital status is included, taking a value of 1 for married individuals and zero otherwise

- Age - to assess for the non-linear relationship between age and wages, the model includes a variable measuring age of the employee and the age square.
- Educational level - three dummy variables are included in the model: a group 2 dummy variable with value of 1 if a person’s highest completed education was secondary or post-secondary vocational education and zero otherwise; group 3 is set to 1 for persons whose highest educational attainment was a completed graduate tertiary degree, zero otherwise; and group 4 represents individuals that have completed post-graduate tertiary degree. The reference category refers to employees who have not completed any schooling or whose highest completed education was primary or lower secondary education.
- Children - In line with previous studies, the model includes a measure for the presence of children, with a binary variable indicating if the household has children under the age of 15.

The following job characteristics are included, given that wage levels have been previously identified as significantly associated with work experience including the number of years with an employer, employment status and occupation (Becker, 2010; Mincer, 1974):

- Number of years with employer - literature suggests that wage levels are associated with work experience, but due to data limitations, the model includes a measure of the age – years of schooling – 6 .
- Employment status - to differentiate working arrangements, a dummy variable is included, equal to 1 for full-time employment and 0 for part-time employment.
- Occupation - five dummy variables are included-for managers, professionals, craft and related trade workers, elementary occupations and other occupations. Service and sales workers is set as the benchmark category.

The following employer characteristics are included, given that wage levels have been previously identified as significantly associated with the characteristics of the broad sector and industry ((Becker, 2010; Borjas, 2005)):

- Sector - seven dummy variables for industry sectors are included: agriculture, manufacturing, construction, public administration, education, health and other sectors, with trade as the benchmark category.

3.3 OLS regression and the Oaxaca-Blinder (1973) decomposition

The analysis is based on a Mincerian earnings function, which is modelled by (Mincer, 1974) and includes experience (as a proxy for the job training), its square, education and other variables that are associated with individuals' earnings. This model is specified for the formal and the informal sector as follows:

$$lnw_i^{FE} = x_i^{FE}\beta + \varepsilon_i^{FE} \quad (1)$$

$$lnw_i^{IE} = x_i^{IE}\beta + \varepsilon_i^{IE} \quad (2)$$

where i = {1, ... , N} denotes individuals and lnw_i^{FE} and lnw_i^{IE} refers to the log of the hourly wage of the individual i in the formal and the informal sectors, respectively. x_i^{FE} and x_i^{IE} represent individual characteristics such as education, experience, gender, location, etc. The estimated earning equations for formal and informal workers are used to calculate the difference in wages between the formal sector and the informal sector.

First, taking the gross wage differential (denoted as W) between the Formal employment (FE) and the informal employment (IE) groups is the difference in the predicted logarithmic daily wages of the two groups (the higher wage group: FE; and the lower wage group: IE)

$$W = E(Y_{FE}) - E(Y_{IE})$$

where E(Y) denotes the expected value of the outcome variable, is accounted for by group differences in the predictors.

Based on the linear model

$$Y_l = X_l^{'}\beta_l + \epsilon_l, \quad E(\epsilon_l) = 0, \quad l \in (FE, IE)$$

where X is a vector containing the predictors and a constant, β contains the slope parameters and the intercept, and ϵ is the error, the mean outcome difference can be expressed as the difference in the linear prediction at the group-specific means of the regressors. That is,

$$W = E(Y_{FE}) - E(Y_{IE}) = E(X_{FE})'\beta_{FE} - E(X_{IE})'\beta_{IE} \quad (3)$$

because

$$E(Y_l) = E(X_l'\beta_l + \epsilon_l) = E(X_l'\beta_l) + E(\epsilon_l) = E(X_l'\beta_l)$$

where $E(\beta_l) = \beta_l$ and $E(\epsilon_l) = 0$ by assumption.

To identify the contribution of group differences in predictors to the overall outcome difference, (3) can be rearranged, for example, as follows (see Winsborough and Dickinson [1971]; Jones and Kelley [1984]; and Daymont and Andrisani [1984]):

$$W = \{E(X_{FE}) - E(X_{IE})\}'\beta_{IE} + E(X_{IE})'(\beta_{FE} - \beta_{IE}) + \{E(X_{FE}) - E(X_{IE})\}'(\beta_{FE} - \beta_{IE}) \quad (4)$$

This is a “threefold” decomposition; that is, the outcome difference is divided into three components:

$$W = E + C + I$$

The first component,

$$E = \{E(X_{FE}) - E(X_{IE})\}'\beta_{IE}$$

amounts to the part of the differential that is due to group differences in the predictors (the “endowments effect”).

The second component,

$$C = E(X_{IE})'(\beta_{FE} - \beta_{IE})$$

measures the contribution of differences in the coefficients (including differences in the intercept). And the third component,

$$I = \{E(X_{FE}) - E(X_{IE})\}'(\beta_{FE} - \beta_{IE})$$

is an interaction term accounting for the fact that differences in endowments and coefficients exist simultaneously between the two groups.

The decomposition shown in (4) is formulated from the viewpoint of group IE. That is, the group differences in the predictors are weighted by the coefficients of group IE to determine the endowments effect (E). The E component measures the expected change in group IE’s mean outcome if group IE had group FE’s predictor levels. Similarly, for the C component (the “coefficients effect”), the differences in coefficients are weighted by group IE’s predictor levels. That is, the C component measures the expected change in group IE’s mean outcome if group IE had group FE’s coefficients.

The main problem in measuring the wage gap between the formal and the informal employment is trying to answer the following counterfactual question: ‘What happens if the allocation of individuals to the informal employment and the formal employment are not random?’ A participation decision of workers to participate in any sector (endogenous selection) brings selectivity bias to occur and affects the formal–informal wage gap. To implement this procedure, a probit regression is run in the first stage to find the probability of working in the informal sector as a function of original variables and an additional identifying variable.

This then leads to a practical difficulty in finding excluded variables for regression analysis. As mentioned above, workers may not be a random sample of working-age population. Unobserved factors may determine workers' sectoral choice. We aim to solve the problem, sample selection, by introducing a correction term obtained from a probit model into the wage equation. Using the available data, we expect the proportion of the informal sector to all workers to influence the sectoral choice of workers without having a (direct) impact on wages.

3.4 Dependent and Independent Variables

Dependent and independent variables with their description are included herewith in this section.

Variable Name	**Description**
Dependent Variable	
Log of hourly wage rate	It includes wage earned from employment.
Independent Variables	
Gender	Female = 1, Male = 0
Experience	Experience is calculated as: age – years of schooling – 6
Experience squared	Square of experience
Education	Categories of education is based upon the levels completed. There are seven categories: 1. Illiterate 2. Below primary 3. Primary 4. Tenth Grade 5. Secondary 6. Bachelor 7. Masters and above
Caste Group	Caste of sample is divided into six categories as below: 1. Khas 2. Janajati 3. Adhibasi 4. Madhesi 5. Dalit 6. Others

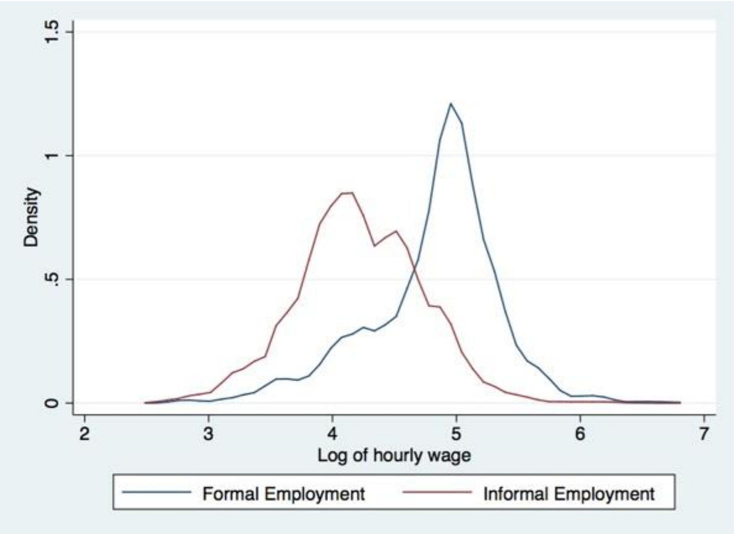
Married Status	Married = 1, Unmarried = 0
Children	Number of children in household
Occupation	Occupation class is based on following five categories: 1. Elementary occupation 2. Plant & machine operators 3. Skilled Agriculture workers 4. Clerical Services 5. Managers, professionals and technicians
Vocational Training	Workers who received vocational training = 1 & who did not = 0
Migration for work	Dummy variables, Migrated for work = 1 Not migrated for work = 0
Overtime	Dummy variable; working hour > 40 hours per week = 1 working hour $\leq$ 40 hours per week = 0
Total Chores	Number of hours spent per day in household activities
Firm Size	Size of the firm based on the number of workers employed 1. Small size firm (< 5) 2. Medium size firm (5 to 20) 3. Large size firm (>20)
Job Sector	Division of job as per the economic sectors: 1. Agriculture 2. Mining and Quarrying; Electricity, gas and water supply 3. Construction 4. Manufacturing 5. Market Service 6. Non-market service 7. Others

4. Results and Data Analysis

In this section, we present the result of our descriptive statistics, result of basic regression analysis and Oaxaca-blinder wage gap decompositions.

4.1 Descriptive statistics

Figure 1 Changes in the distribution of Log Real Hourly Wages, by Formal and Informal Employment



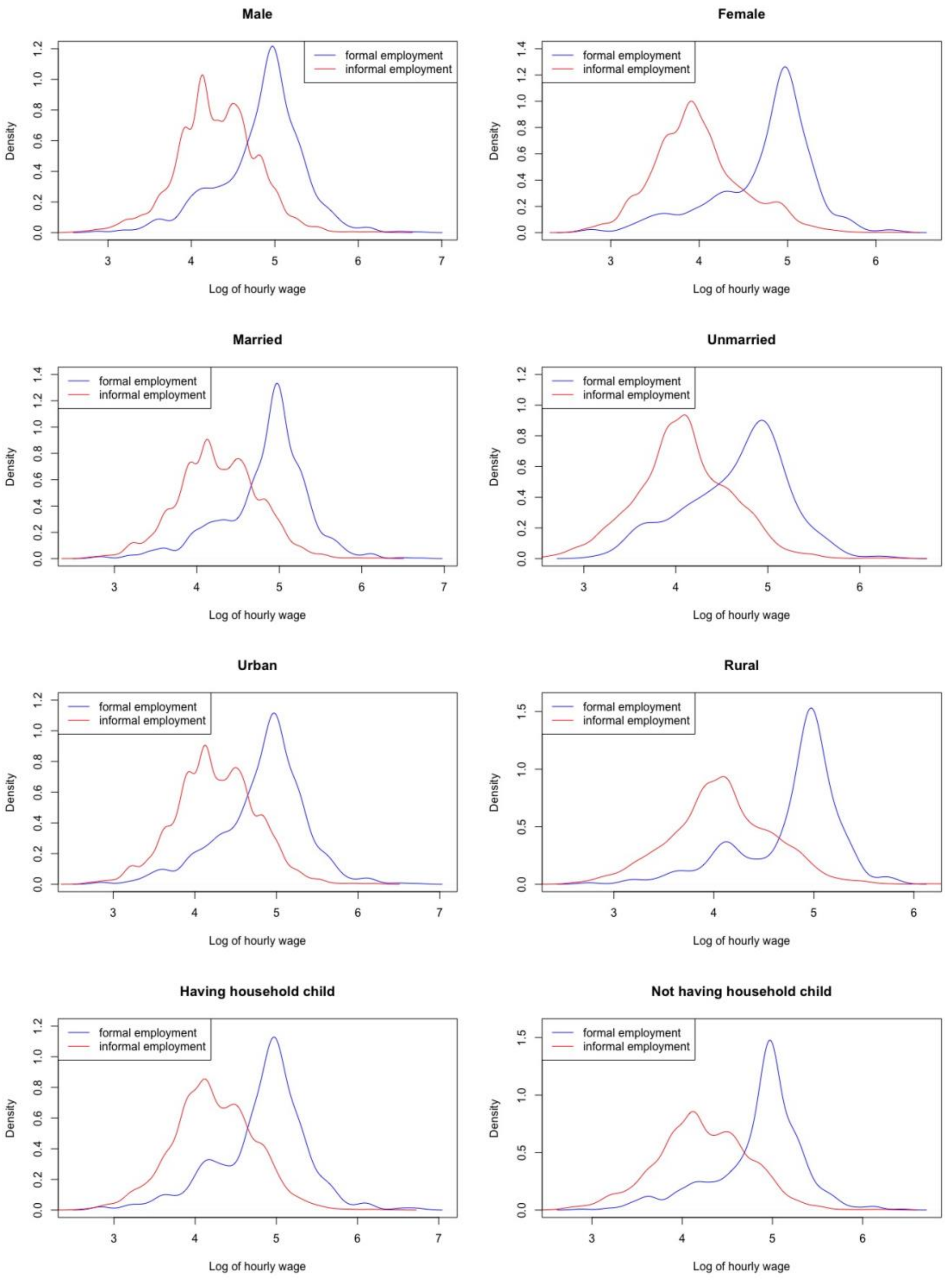
Source: Authors' compilation based on NLFS III, CBS (2018)

Figure 1 shows that the earnings distribution is shifting towards the left for informal workers compared to formal workers and shows much greater variation. As per our data from table 9, employees who are in informal employment and are female, based in urban area, do not have vocational training, have not migrated for work, does overtime work, and have children earn on average less than respective attributes in formal employment. The kernel density curve below shows the variations across formal and informal employment based on these individual and household characteristics.

Table 1 Average real daily wage rate for formal and informal employment by personal characteristics

	Formal employment	Informal employment	Pooled	PRG
Male	141.42	82.79	90.75	41.46
Female	131.42	61.80	70.22	52.98
Rural	131.09	77.71	82.78	40.72
Urban	141.57	77.00	86.73	45.61
Vocational training	149.64	88.48	102.54	40.87
No vocational training	135.20	75.49	82.32	44.16
Migrated for work	162.78	89.66	107.87	44.92
Not migrated for work	132.16	75.83	82.37	42.62
Overtime	113.04	73.86	77.35	34.66
No overtime	160.38	85.17	101.53	46.89
Children	141.51	78.06	86.38	44.84
No children	142.02	78.30	89.29	44.87

Figure 2 Changes in the distribution of Log Real Hourly Wages, by Formal and Informal Employment across household characteristics



Male
Female
Married
Unmarried
Urban
Rural
Having household child
Not having household child
formal employment
informal employment
Density
Log of hourly wage

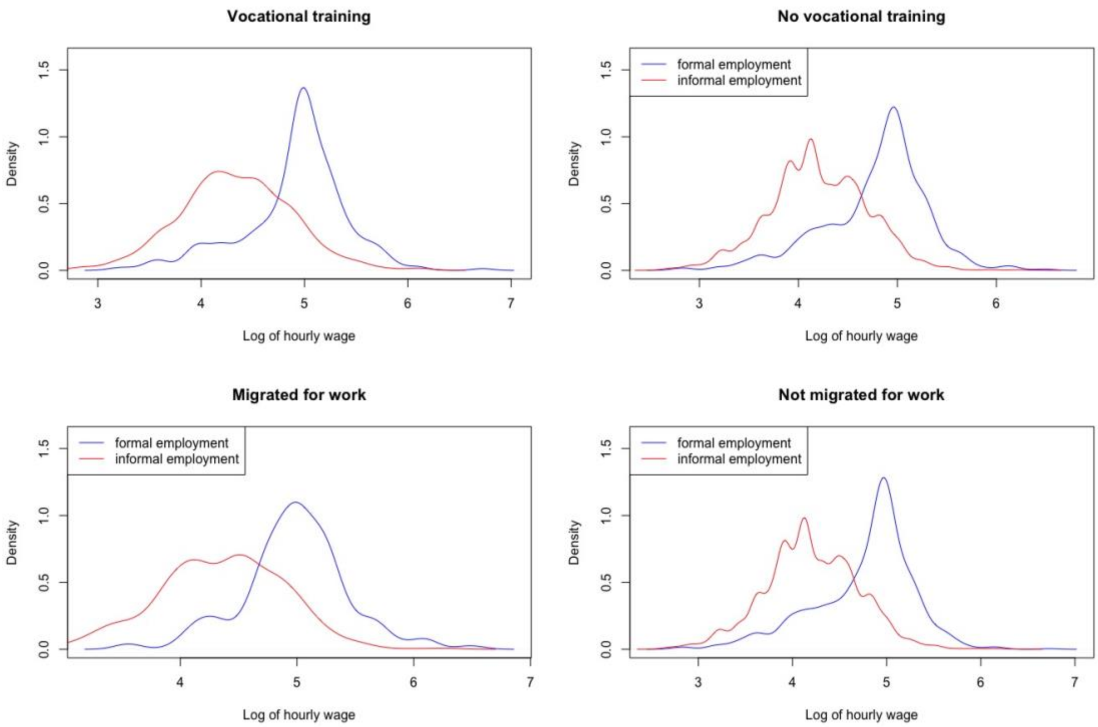


Table 2 Composition of formal and informal employment by education attainment level

	formal employment	Informal employment	Pooled
Illiterate	4.50%	22.70%	20.30%
Below primary	3.70%	17.00%	15.20%
Primary	11.10%	27.40%	25.20%
Up to tenth grade	20.90%	14.60%	15.50%
Secondary	21.90%	10.20%	11.80%
Bachelors	22.00%	5.80%	7.90%
Masters & above	15.80%	2.30%	4.10%

As per our data, the proportion of employees in formal employment increases with the increase in the educational attainment level. Table …. shows that the highest proportion of employees of our sample data for formal employment are of bachelor level at 22% and for informal employment are of Primary level with 27.4%. Employees of primary level of education and below constitutes of 67.10 % of total employees in informal employment. Employees of bachelors and above makes for 37.80% of formal employment. There appears an inverse relationship of education among formal and informal employment, where lower attainment of education pushes labor force towards informal employment on average and subsequently formal employment as the education level increases.

The average real hourly wage rate for both of the formal and informal employment increases with the increase in education level, aligning with the preexisting literature of human capital theory. Illiterate workers on an average earn Rs. 64.07 per hour, the lowest among education level, and masters and above workers earn Rs. 202. 654, the highest among education level for formal employment. Similarly, Illiterate workers on an average earn Rs. 64.01 per hour, the lowest among education level, and masters and above workers earn Rs. 159.09 per hour, the highest among education level for informal employment. Although the increase in education attainment level increases the average real hourly wage rate for all employment, the percentage real gap highly exists across all level. The wage gap is highest for tenth grade at 38.67% and secondary level at 38.34% and starts decreasing for bachelor level at 30.60% and masters and above level at 21.49%. There is not much significant difference of real hourly wage rate for Illiterate and below primary level workers, with the lack of human capital forcing workers to earn the lowest amount. The fig in ….. shows the kernel density curve of log hourly wage for all the education level in formal and informal employment, which supports the preceding conclusion on the existing wage gap.

Table 3 Average real hourly wage rate for formal and informal employment by education attainment level

	formal employment	**Informal employment**	**Pooled**	**PRG**
Illiterate	64.07	64.011	64.01	0.09
Below primary	77.93	74.728	74.83	4.11
Primary	89.63	74.544	75.41	16.83
Tenth Grade	120.17	73.69	81.98	38.67
Secondary	134.283	82.796	95.44	38.34
Bachelors	166.662	115.66	134.38	30.60
Masters & above	202.654	159.098	181.17	21.49

Figure 3 Changes in the distribution of Log Real Hourly Wages, by Formal and Informal Employment across educational level

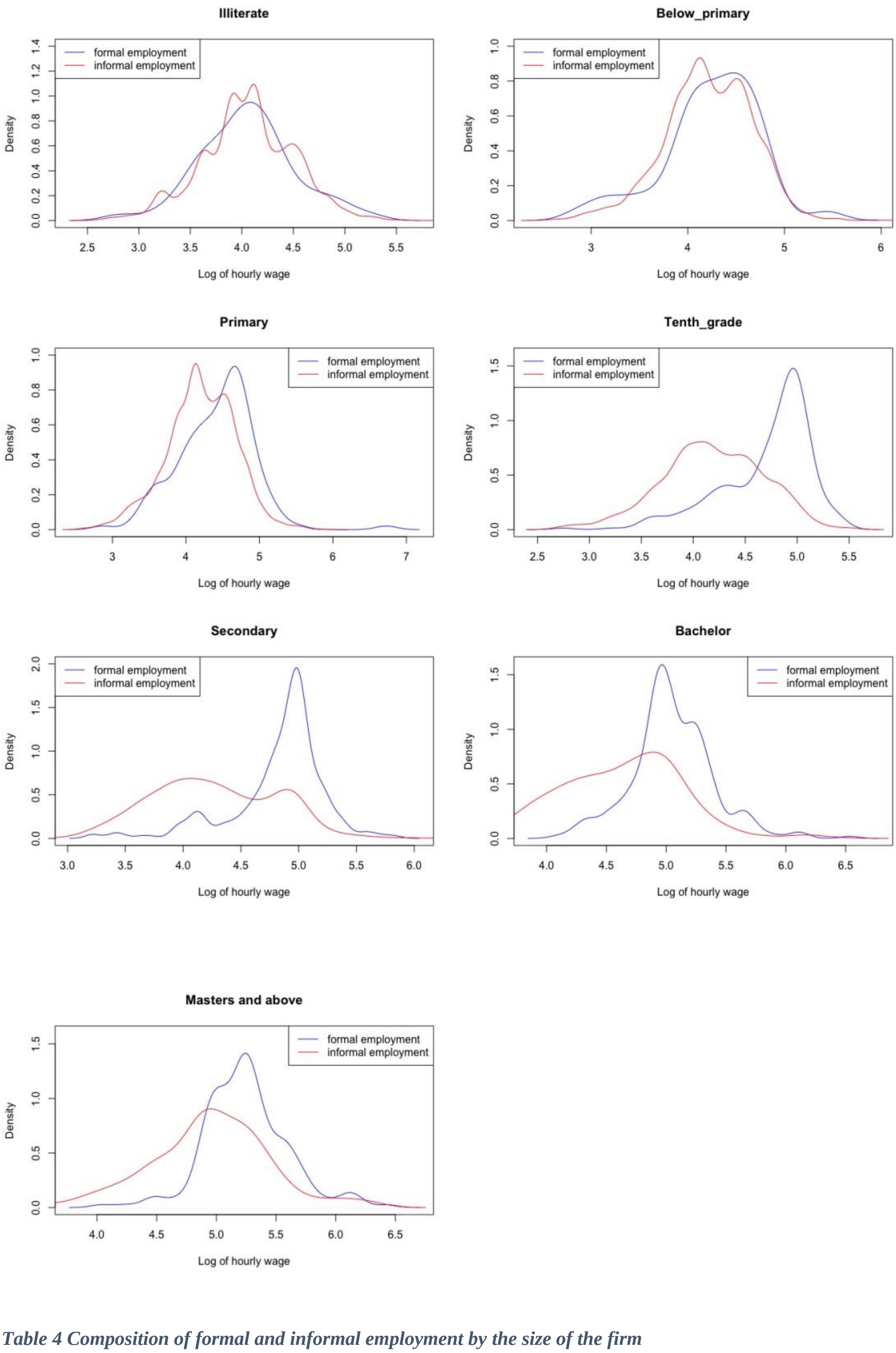


Table 4 Composition of formal and informal employment by the size of the firm

	Formal employment	Informal employment	Pooled
Small size firm	4.60%	49.70%	43.80%

Medium size firm	3.10%	24.70%	21.90%
Large size firm	92.30%	25.50%	34.30%

As per our data, small size of firm have the highest share at 43.80% and medium size firm have lowest at 21.90%, while large size firm composes of 34.30%. However, there has been an uneven distribution of employment on the basis of size of the firm, where large size of firm makes for 92.30% of total employment. Small and medium size firm are visibly lower at 4.60% and 3.10% respectively. This is not a surprise though, considering the fact that the definition of formal and informal employment in this study is based on attainment of job based benefits including temporary leave, leave based benefits, contract of job, social security contributions, sickness, illness & injury based compensations as well as registered to the authority which is legally mandatory by the authority and implemented by the large size firms, whereas medium and small size firm lacks behind on ensuring these benefits. It is interesting to note half of the informal employment at the small size firm, which is evident to the nature of the jobs being carried out in the informal employment in Nepal such as street vendors, domestic workers, construction workers, transportation workers and agriculture workers.

The average hourly wage rate for formal and informal employment both increases with the increase in the size of the firm, as evident in the above table   The average hourly wage rate is highest at Rs. 144.41 for large size firm and lowest at Rs. 74.44 in the formal employment. Although, large size firm have highest earning at Rs. 88.04 and small size firm with lowest earning at Rs. 71.90, this is a significant drop of earning earned in the informal employment over formal employment. The increase in the size of the firm also results in the increase of average wage gap as seen from the table, where a minimal gap of 3.41 % exists in small size firm and staggering 39.03 % in large size firm, which highlights the essence of formalization of informal employment to formal employment.

Table 5 Average real hourly wage rate for formal and informal employment by size of the firm

	Formal employment	**Informal employment**	**PRG**
Small size firm	74.44	71.90	3.41
Medium size firm	96.85	76.85	20.65
Large size firm	144.41	88.04	39.03

Figure 4 Changes in the distribution of Log Real Hourly Wages, by Formal and Informal Employment across size of the firm

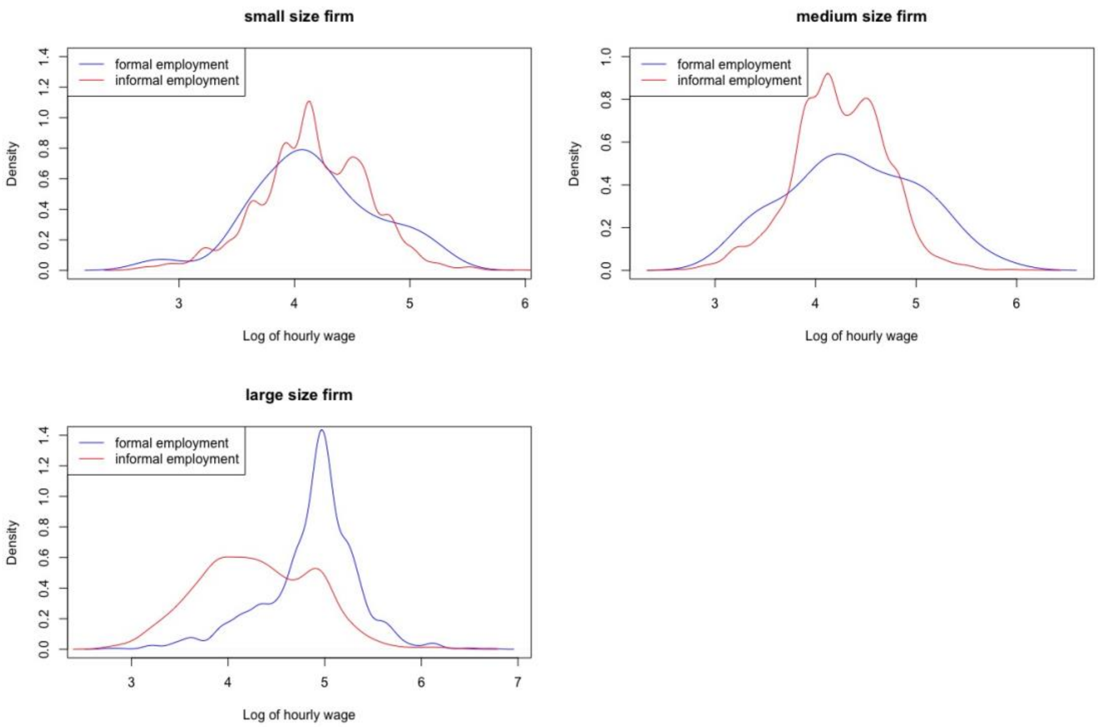


Table 6 Composition of formal and informal employment by the workers place of work

	Formal employment	Informal employment	Pooled
Government	67.80%	10.30%	17.90%
Private Institution	24.50%	21.00%	21.50%
Private Business	2.80%	46.60%	40.80%
Others	4.90%	22.10%	19.80%

As observed from table 11, despite private business making for 40.80% of our total share of employment, 99% of them are observed operating in informal employment and there is existence of 7.41% wage gap. Government sector has the highest share at 67.80% of total formal employment and on average earns highest hourly wage rate across place of work, with an existence of 34.94% wage gap. Similarly, private institutions makes for 21.50% of total employment and is evenly distributed across formal and informal employment, and there exists a wage gap of 38.24%. The rest of the place of work, largely are on informal employment. Private business have the lowest wage gap among the place of work at 7.41% and the rest of the work place mandates the policy discourse regarding formalization of them towards formal employment in Nepal.

Table 7 Average real hourly wage rate for formal and informal employment by the workers place of work

	Formal employment	Informal employment	Pooled	PRG
Government	147.69	96.08	121.87	34.94
Private Institution	120.23	75.95	82.62	36.83
Private Business	80.67	74.69	74.74	7.41
Others	146.52	75.08	77.39	48.76

Figure 5 Changes in the distribution of Log Real Hourly Wages, by Formal and Informal Employment across worker's place of work

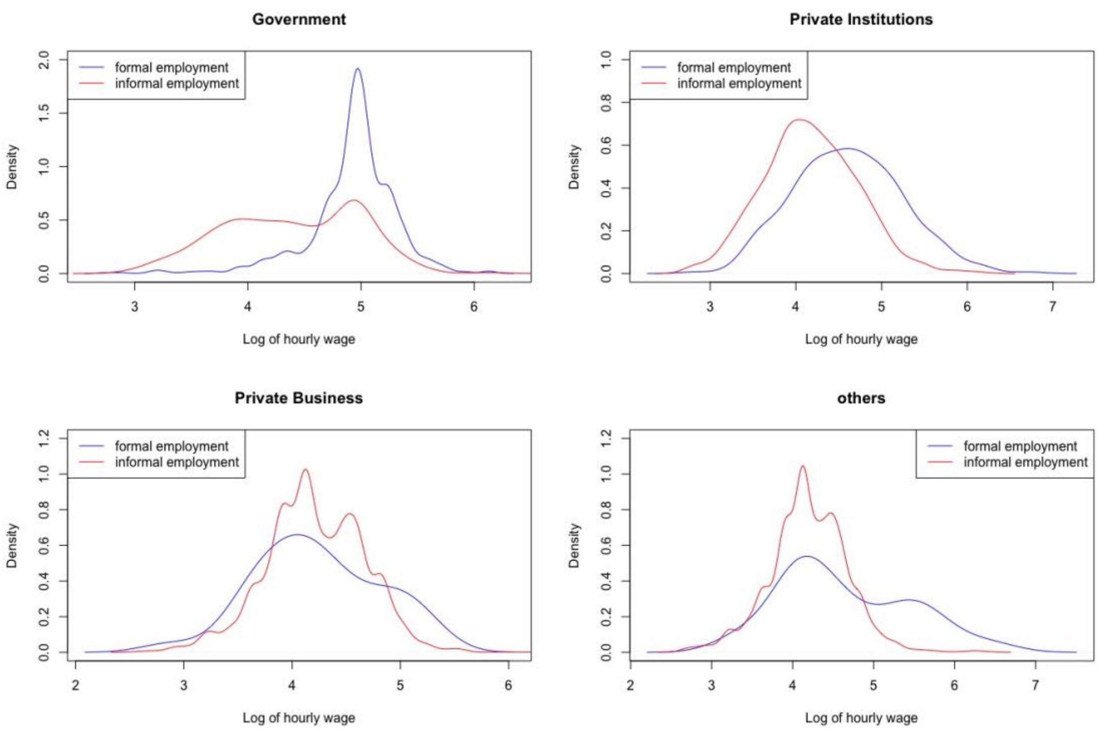


Table 8 Composition of formal and informal employment by the division of job as per the economic sectors

	Formal employment	Informal employment	Pooled
Agriculture	8.00%	13.60%	12.90%
Mining and Quarrying; Electricity, gas and water supply	3.40%	2.90%	3.00%
Construction	1.30%	32.90%	28.70%
Manufacturing	7.20%	13.90%	13.00%
Market Service	20.90%	18.70%	19.00%
Non-market service	57.90%	15.80%	21.30%
Arts, entertainment and other service activities	1.30%	2.20%	2.10%

As per the table……..construction sector at 28.70% makes for the highest share of total employment and informal employment. Considering the fact that, there is only 1.30 % employment on construction sector formally, and a wage gap of 49.5% ; most among any economic sector, workers on this sector have been subject of economic, work place and safety based vulnerabilities. Non market service makes for 21.30% of total employment, and 57.90% of the formal employment. Despite having the highest average wage earned among any economic sectors within informal employment at Rs. 91.56 and second highest average wage earned within formal employment at Rs. 149.59, there exists 38.79 % of wage gap in the non-market service.

All of the economic sectors have high degree of wage gap as seen on table 12. Market service has 48.48% wage gap, Mining and Quarrying; Electricity, gas and water supply at 47.41% , Arts and entertainment and other service activities at 47.55% . The lowest wage gap is at 28.89% for manufacturing sector and that itself reflects the precarious situation among formal and informal employment within the division of the job as per the economic sectors. There exists a policy discourse on formalizing the jobs as per economic sectors, where each of the different sectors should be studied separately on the basis of the socio-economic significance each of the sector makes in Nepalese labor market.

Table 9 Average real daily wage rate for formal and informal employment by the division of job as per the economic sectors

	Formal employment	Informal employment	Pooled	PRG
Agriculture	74.34	52.02	53.85	30.02
Electricity, gas and water supply	136.48	71.77	81.48	47.41
Construction	167.36	84.52	85.02	49.5
Manufacturing	102.59	72.95	75.12	28.89
Market Service	145.17	74.79	85.01	48.48
Non-market service	149.59	91.56	112.33	38.79
Arts, entertainment and other service activities	146.71	76.95	82.74	47.55

Figure 6 Changes in the distribution of Log Real Hourly Wages, by Formal and Informal Employment across economic job sectors

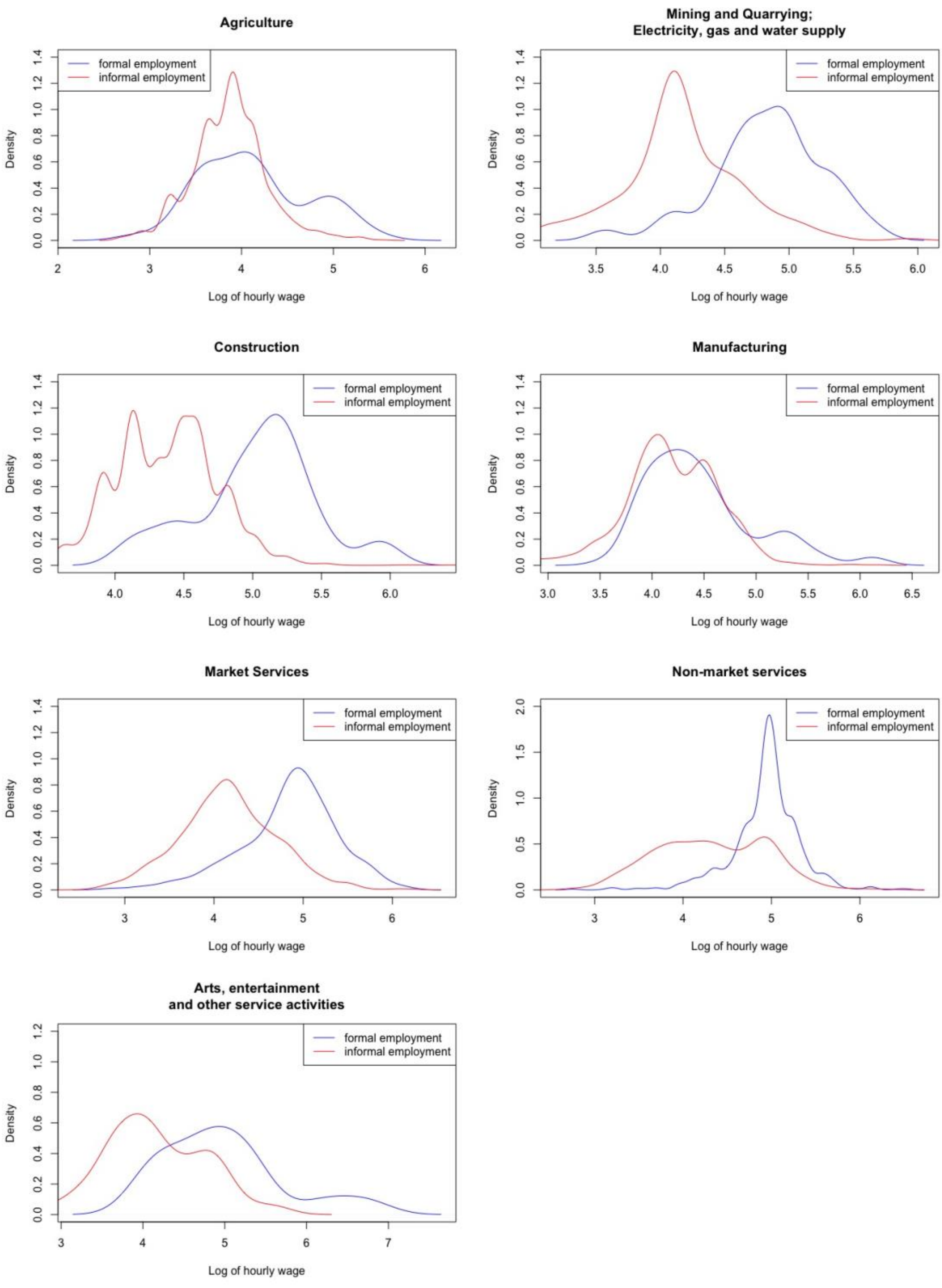


Table 10 Average real daily wage rate for formal and informal employment by the caste group

	Formal employment	Informal employment	Pooled	PRG
Khas	150.83	87.94	102.99	41.70
Janajati	133.48	80.01	86.22	40.06
Adhibasi	103.62	66.36	68.45	35.96

Madhesi	121.4	72.31	78.55	40.44
Dalit	99.42	70.21	71.17	29.38
Others	143.14	66.92	70.39	53.25

As per the table 14, of our total sample, Khas makes for 30.30% , Janajati at 28.10% , Dalit at 14.60%, Adhibasi at 13.20 %, Madhesi at 10.20% and rest . Over half of total employed in formal employment are Khas at 55.00 % followed with 24.70% of Janajati, 9.80% for Madesi . Dalit groups appears to have administrative and societal barriers to enter the formal employment with only 3.60 % of formal employment being Dalits. Of the total informal employment, Janajati has the highest share at 28.60 %, followed with 26.50% of Khas, 16.40 % Dalit, 14.40% Adhibasi and rest of others.

As per table 13, Khas group earns the highest average hourly wage in both formal and informal employment at Rs. 150.83 and Rs. 87.94 respectively, and subsequently highest wage gap exists among any caste group at 41.70% if we ignore the ambiguity of *others* caste group. There is a clear pattern here among the average hourly wage earning on caste group for formal and informal employment, as well as the existence of the wage gap too, with Khas group followed by Janajati and Madhesi. The average hourly wage for Janajati is Rs. 133.48 and Rs. 80.01 for formal and informal employment respectively and there exists wage gap of 40.06 %. Similarly, the average hourly wage for Madhesi is Rs. 121.4 and Rs. 72.31 for formal and informal employment respectively with a wage gap of 40.44 % . Dalit group have the lowest of average hourly wage rate earnings across both formal and informal employment at Rs. 99.42 and Rs. 70.21 respectively with an existence of 29.38 % wage gap; lowest among any caste group.

Table 11 Composition of formal and informal employment by the caste group

	Formal employment	Informal employment	Pooled
Khas	55.00%	26.50%	30.30%
Janajati	24.70%	28.60%	28.10%
Adhibasi	5.60%	14.40%	13.30%
Madhesi	9.80%	10.20%	10.20%
Dalit	3.60%	16.40%	14.60%
Others	1.20%	3.90%	3.50%

Figure 7 Changes in the distribution of Log Real Hourly Wages, by Formal and Informal Employment across caste group

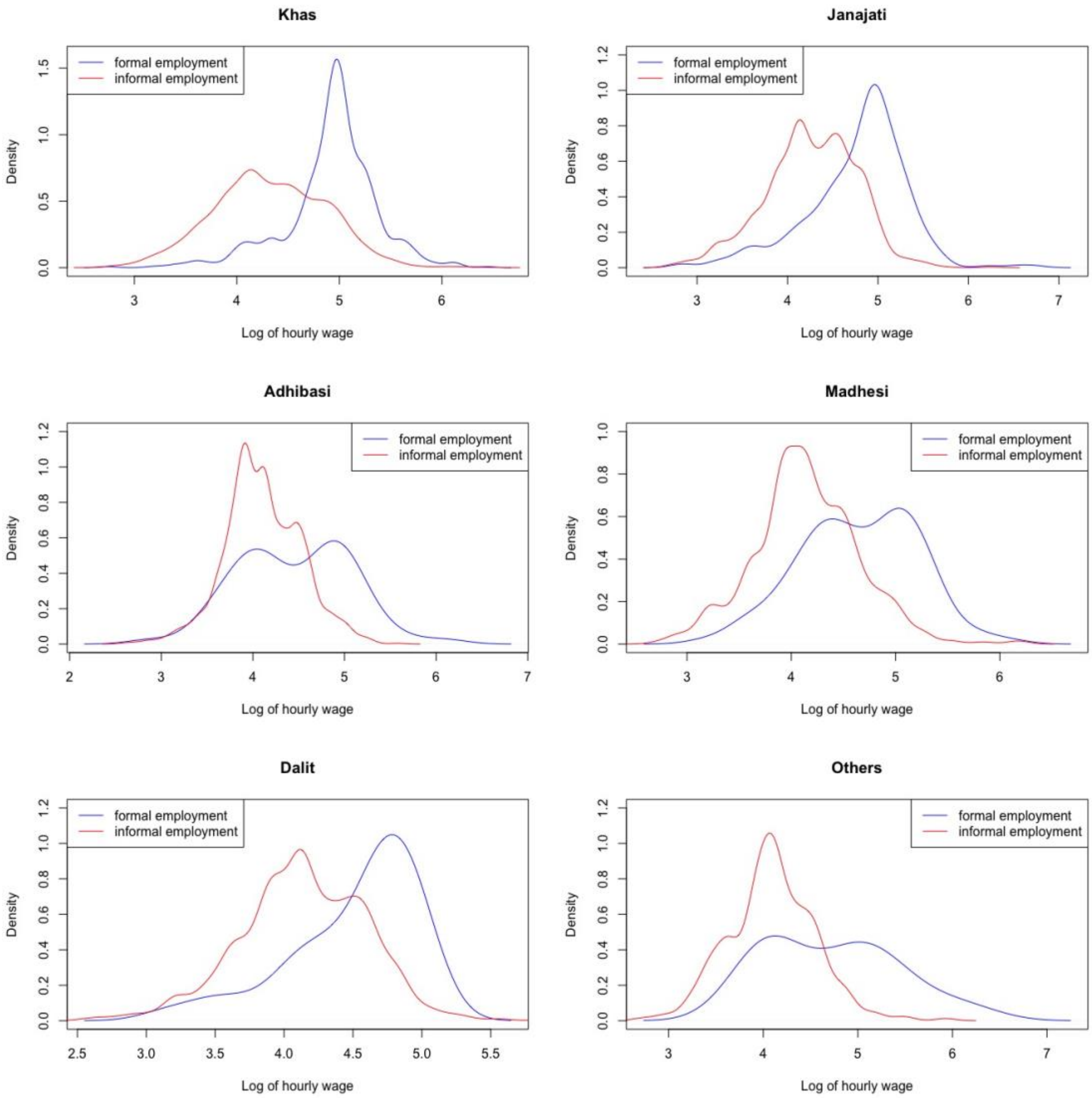


Table 12 Average real daily wage rate for formal and informal employment by the occupation class

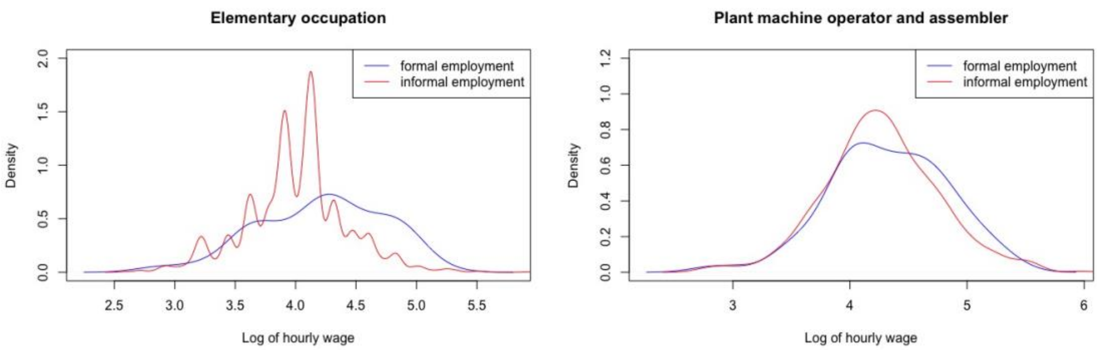
	Formal employment	Informal employment	Pooled	PRG
Elementary occupation	76.68	59.76	60.33	22.07
Plant machine operator and assembler	85.2	80.38	80.81	5.66
Skilled agriculture and trade workers	79.3	85.69	85.59	-8.06
Clerical, service and sales workers	115.68	71.45	81.70	38.23
Managers, professional and technicians	162.85	106.31	128.48	34.72

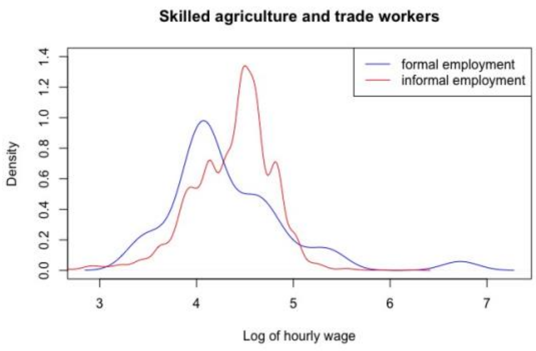
Table …. shows that elementary occupation makes for 32.80%, followed with skilled agriculture and trade workers at 26.60% and managers, professional and technicians at 20.80% of total employment. Over the formal employment, managers, professional and technicians have the highest share at 61.90% followed with clerical service and sales workers at 21.80%. The trend is completely opposite in informal employment where 36.60% are of elementary occupation, followed with skilled agriculture and trade workers at 30.20% and managers, professionals and technicians at 14.60%. Skilled agriculture and trade workers despite having second highest share of total employment makes for only 3.20% of formal employment, and has the only negative wage gap, where workers in informal employment earns 8.06% more than formal employment. This is do more with the lack of formalization of skilled agriculture workers and trade workers than highest average earnings over occupation class as seen from table …….. Clerical, service and sales workers have the highest wage gap at 38.23 % followed with Managers, professionals and technicians having the wage gap of 34.72 % between formal and informal employment.

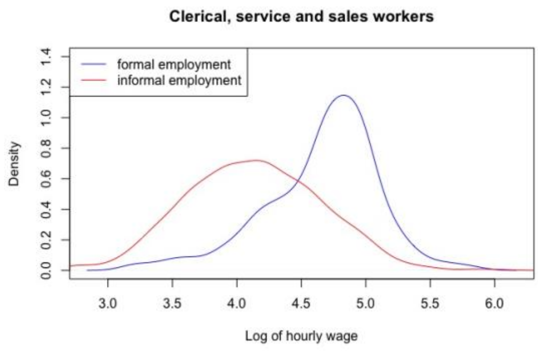
Table 13 Composition of formal and informal employment by the occupation class

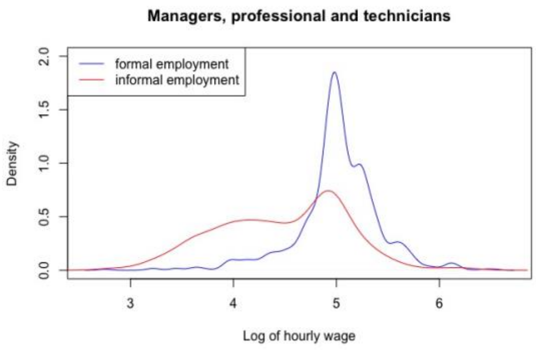
	Formal employment	Informal employment	Pooled
Elementary occupation	8.20%	36.60%	32.80%
Plant machine operator and assembler	4.90%	7.70%	7.30%
Skilled agriculture and trade workers	3.20%	30.20%	26.60%
Clerical, service and sales workers	21.80%	11.00%	12.40%
Managers, professional and technicians	61.90%	14.60%	20.80%

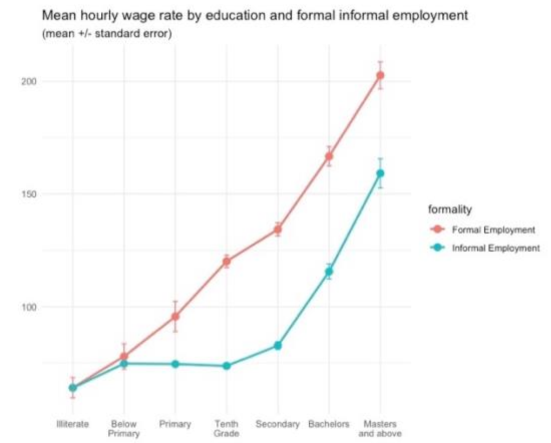
Figure 8 Changes in the distribution of Log Real Hourly Wages, by Formal and Informal Employment across occupation class

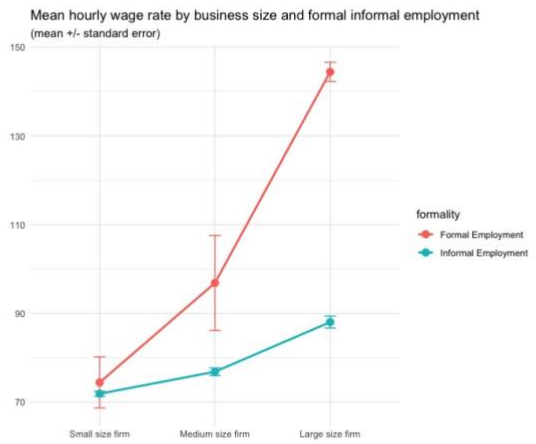












4.2 Basic regression results

We start by using a basic specification and include the control variables individually to gain insights in the relation between them and our variable of interest, namely ‘log hourly wage’. The dependent variable in all the models is the natural log of the hourly wage. In the basic specification our controls for individual and household characteristics are experience, experience squared, educational attainment, caste group, married status and having household children. In order to allow for a nonlinear relationship between experience and our dependent variable, we include both experience and experience squared. Table ….. presents a series of regressions in which we add controls at the occupational level (job-related characteristics) to the basic specification. In the regressions contained in Table …… we have also added employer related controls. We will now discuss our results.

4.2.1 Control on the individual & household characteristics

In the table 1 and 2, five model specifications of individual and household characteristics are presented for formal and informal employment respectively. The first model is the basic specification of experience, experience squared and education. All the coefficients are significant at 1 percent level, and models except for experience squared are positive. Since the dependent variable is measured in terms of natural logarithm, we can interpret coefficients of the independent variables in terms of percental changes. For all the dummy variables, we calculate coefficient -1 in order to obtain the correct coefficient in the determination of the relationship with the log hourly wage. The signs in the first model is positive for experience and negative for experience squared as predicted by the human capital theory. Education is a dummy variable consisting of seven dummies, each one representing particular education level. The illiterate level is omitted from the analysis and therefore is set as the benchmark category. Those who can simply read and write i.e. below primary level adds 32 percent to the wage rate in comparison to illiterate workers in formal employment, whereas 23 percent in informal employment. Primary level adds 55 percent and 32 percent respectively, tenth grade level adds 91 percent and 32 percent respectively, while a secondary level adds 110 percent and 49 percent respectively in formal and informal employment. Bachelor level adds 136 percent in formal employment whereas 85 percent in informal employment. Masters and above level have highest addition to the wage rate for both formal and informal employment at 155 and 113 percent respectively. Although there is positive effect of educational level in informal employment, the addition is greater in formal employment across all levels.

The second column in table 1 and 2 includes a control for gender, with male set as the base category. Gender is not significant for formal employment whereas it is significant at 1 percent significance level for informal employment. Being a male or female in the formal employment does not matter as such, you have similar experience and educational level. Being a female in informal employment decreases average hourly wage rate by 30 percent in comparison to male. There are basic alterations in the magnitude of the coefficients for both formal and informal employment in comparison to basic models, but the nature and significance of different coefficients remains the same.

The third column in table 1 and 2 includes a control for caste group consisting of five dummies, each of those representing different caste group in Nepal. Caste group *Khas* is set as the benchmark category. There is no significance of caste group across all levels in formal employment. In informal employment, *Janajati* group is significant at 10 percent significance level, and the rest of the caste group are significant at 1 percent significance level. Being *Janajati* in average earns 2 percent more in comparison to *Khas*, whereas *Adhibasi* earns 6 percent less, *Madhesi* caste earns 9 percent less, *Dalit* earns 5 percent and *other* caste group earns 10 percent in informal employment. The nature and significance of different coefficients with control for caste group remains the same, while there is change in the magnitude of the coefficients.

The fourth column in table 1 and 2 includes a control for married status, with unmarried set as the benchmark category for formal and informal employment respectively. Marital status is not significant for formal employment, while it is significant at 1 percent level for informal employment. Being married earns 9 percent more hourly wage rate in average in comparison to unmarried in informal employment. The nature and significance of different coefficients with control for marital status remains the same, while there is slight change in the magnitude of the coefficients.

The fifth column in table 1 and 2 includes a control for having a household child, with not having children set as the benchmark category for formal and informal employment respectively. Having household children is significant for informal employment at 5 percent significant level while it is not significant for formal employment. Having a household child reduces average hourly wage rate by 1 percent in comparison to not having children. The nature and significance of different coefficients with control for household child remains the same across all kinds of employment, while there is slight change in the magnitude of the coefficients.

4.2.2 Control on the job characteristics

In table 3 and 4, we present a series of specifications for formal and informal employment in which, besides controls at the individual and household characteristics level, we also include controls at the job characteristics. For the sake of comparison, in column 1 of table 3 and 4, we reproduce the same model specifications as in column 5 of table 1 and 2 respectively. The following six columns summarize the effect of adding one control variable on the job characteristics to each specification. Hence, the last model specification includes all control variables, both on the individual and the job characteristics.

We start by adding occupation class control in the column 2 for table 3 and 4, consisting of 5 dummies, each representing the occupation group. The omitted occupation class is "Elementary occupation". Plant, machine operators and assembler and skilled agriculture and trade workers are not significant in formal employment, whereas they are significant at 1 percent significance level in informal employment. Similarly, both the clerical, service and sales workers and managers, professionals and technicians are significant at 1 percent significant level for formal and informal employment. In the formal employment, being a clerical, service and sales workers adds 20 percent more to the hourly wage rate, whereas managers, professionals and technicians adds 38 percent more in comparison to elementary occupation. Similarly, in the informal employment, being plant, machine operators and assemblers adds 9 percent, skilled agriculture and trade workers adds 24 percent, managers, professionals and technicians adds 15 percent whereas being a clerical, service and sales workers decreases the average wage rate by 5 percent in comparison to the elementary occupation. The interesting effect is seen on being a clerical, service and sales workers, which adds to the wage rate in formal employment, but decreases the wage rate in the informal employment.

The third column in table 3 and 4 consists of control for the vocational training, where not having vocational training is set as the benchmark category. It appears vocational training is not significant for the formal employment, whereas it is significant at 5 percent significance level for the informal employment. Workers who have vocational training in average adds 2 percent to the wage rate in comparison to the ones who do not have any form of vocational training in the informal employment.

The fourth column in the table 3 and 4 includes of control for workers who have left their hometown, migrated to new places in search of the job, with those who were not migrated set at the benchmark. Migrated for work is significant at 1 percent for both the formal and informal employment. Workers who are migrated adds 11 percent in formal employment, and 6 percent in informal employment to the average wage rate in comparison to those who are not migrated for work.

The fifth column in the table 3 and 4 includes control for total chores per hour done in the formal and informal employment respectively, where workers who did not do any kind of chores are set at the baseline. It is significant at 5 percent in both the formal and informal employment, although the effect is positive on formal employment and negative on informal employment. Workers who does household chores adds 1 percent more in the formal employment, whereas they reduce 1 percent in informal employment of the average wage rate in comparison to those who don't do any household chores.

The sixth column in table 3 and 4 includes a control for workers who does overwork, which is calculated as workers who work more than 40 hours per week in formal and informal employment respectively. Overtime is significant at 1 percent for both the formal and informal employment. Workers who does overtime earns 23 percent in formal employment and 8 percent less in the informal employment in comparison to those who do not do any kind of overtime work.

Including the control for the occupation class makes the *Adhibasi* caste group significant at 10 percent significance and married status significant at 10 percent significance level from the original model in the formal employment. Similarly, control for the occupation class reduces the significance of *Janajati* caste group from 5 percent significance level to not significant at all in the informal employment. With the control for the migration for work, there is no change in the significance in the formal employment, but reduces the significance of household child from 10 percent significance level to no significance level in the informal employment.

Including the control for the migration for work reduces the significance of married status from 10 percent to no significance level and increases the significance of household child from no significance to significant at 10 percent in the formal employment. The control for migration for work has no changes on marital status and household child, but reduces the significance of vocational training from 5 percent significance to 10 percent significance level in the informal employment. Similarly, the control for overtime makes marital status significant from no significance to 5 percent significance level, and household child more significant from 10 percent significance to 1 percent significance in the formal employment. There is not much changes on significance level due to the control for overtime in the informal employment.

4.2.3 Control on the employer characteristics

In table 5 and 6, we present a series of specifications for formal and informal employment in which, besides controls at the individual, household and job-related characteristics level, we also include controls at the employer characteristics. For the sake of comparison, in column 1 of table 5 and 6, we reproduce the same model specifications as in column 6 of table 3 and 4 respectively. The following three columns summarize the effect of adding one control variable on the employer characteristics to each specification. Hence, the last model specification includes all control variables, both on the individual, household, job and employer characteristics.

We start by adding the control on the size of business in the column 2 of table 5 and 6 for formal and informal employment respectively. Size of the firm is a dummy variable consisting of 3 dummies and "small size firm" is set as the baseline category. The medium and large size of the firm is significant at 1 percent significance level for formal employment, whereas only the medium firm is significant at 1 percent and large size firm is not significant for the informal employment. In comparison to small size firm, being in the medium size firm adds 29 percent more to the average hourly wage rate for formal employment and 8 percent more for informal employment. Similarly, being in large size firm adds 31 percent in comparison to the small size firm for formal employment. Since the large size firm are subject to higher legal and administrative scrutiny, and most of them fulfil the criterions of formality, there is no significance of large size firm in the informal employment.

The column 3 of table 5 and 6 includes the control for the economic job sector in formal and informal employment respectively. Job sector is a dummy variable consisting of 7 dummies and "Agriculture" is set as the baseline category. All of the economic job sectors are significant at 1 percent significance level except for arts, entertainment and other service activities in the formal employment which is not significant at all. Similarly, all the economic job sectors are significant at 1 percent significance level except for non-market services in the informal employment, which is not significant at all. Being in the mining and quarrying: electricity, gas and water supply job sector adds 26 percent, construction adds 44 percent, manufacturing adds

18.4 percent, market services adds 25.3 percent and non-market services adds 28.1 percent more in comparison to the agriculture sector in the formal employment. Similarly, being in the mining and quarrying: electricity, gas and water supply job sector adds 12.7 percent, construction adds 33.6 percent, manufacturing adds 11.5 percent, market services adds 8 percent and arts, entertainment and other service activities adds 8.8 percent more in comparison to the agriculture sector in the informal employment.

Including the control for the size of the firm, the significance of plant, machine operators and assemblers increases from 10 percent to 1 percent significance, whereas the significance of Dalit caste increases to 10 percent significance level in the formal employment. Moreover, it also decreases the significance of clerical, service and sales workers from 1 percent significance to 5 percent significance in the informal employment.

Including the control for the job sector seems to make married status less significant from 5 percent to 10 percent in the formal employment, whereas the significance is unchanged in informal employment. Similarly, the control for job sector makes hourly wage rate for below primary education level workers completely insignificant from 1 percent significance level in the formal employment, whereas the significance is similar but there is decrease in the coefficients for informal employment. Furthermore, the control for job sectors increases the significance of large firm to 10 percent significance, whereas reduces the significance of being *Dalit* from 1 percent significance to 5 percent significance, clerical, service and sales workers from 5 percent significance level to no significance at all and total chores per hour from 1 percent significance to no significance in the informal employment.

4.3 Oaxaca- Blinder Decomposition

The overall result of the wage decomposition results by formality in employment for the total sample are presented in the table below:

Table 14 Threefold Blinder-Oaxaca Decomposition in mean wages for formal and informal employment

Average gross hourly wages in Nepalese Rupees (NPR)	
Formal employment	128.81
Informal employment	67.4
difference	0.6476526***
Oaxaca-Blinder effect decomposition	

Endowments (E)	0.3049586***
Coefficients (C)	0.1119687*
Interaction (I)	0.2307253***
Share to total wage gap (%)	
Endowment effect	47.09
Coefficients effect	17.29
Interaction effect	35.62

****, **, * Significant at 1%, 5% and 10% levels, respectively*
Source: Author's own calculations

Group 1 consists of 1153 employees in formal employment while group 2 consists of 7590 employees in informal employment. The mean log hourly wage for the employees in formal employment is 4.86, while the mean log hourly wage for the employees in informal employment is 4.21. As the overall wage difference presents, employees in informal employment earn about 64.77 percent less than in formal employment do. This mean wage difference is next decomposed in three parts. The results of this decomposition can be found in the last three rows of table 6.

Figure 9 Threefold Blinder-Oaxaca Decomposition plot in mean wages for formal and informal employment across endowments

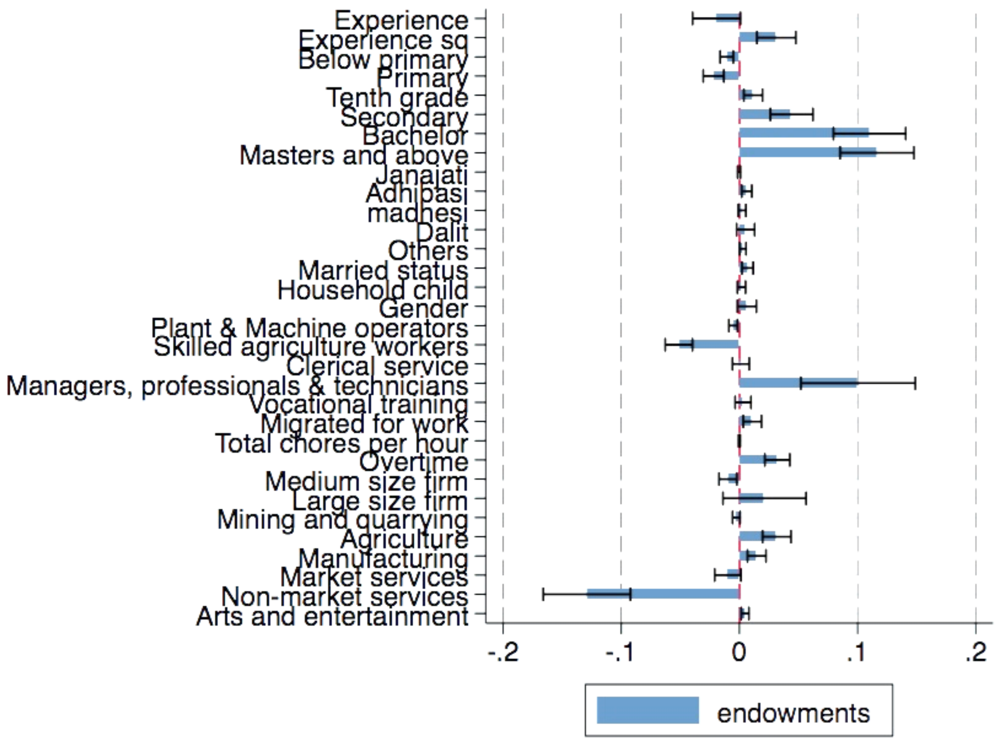


Figure 10 Threefold Blinder-Oaxaca Decomposition plot in mean wages for formal and informal employment across coefficients

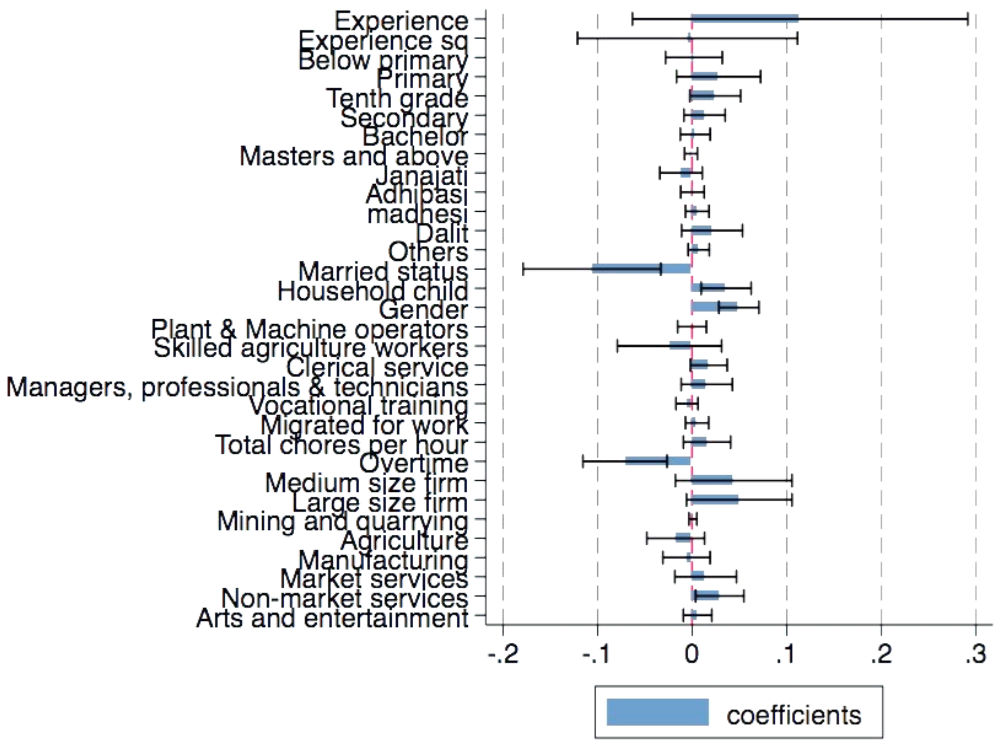
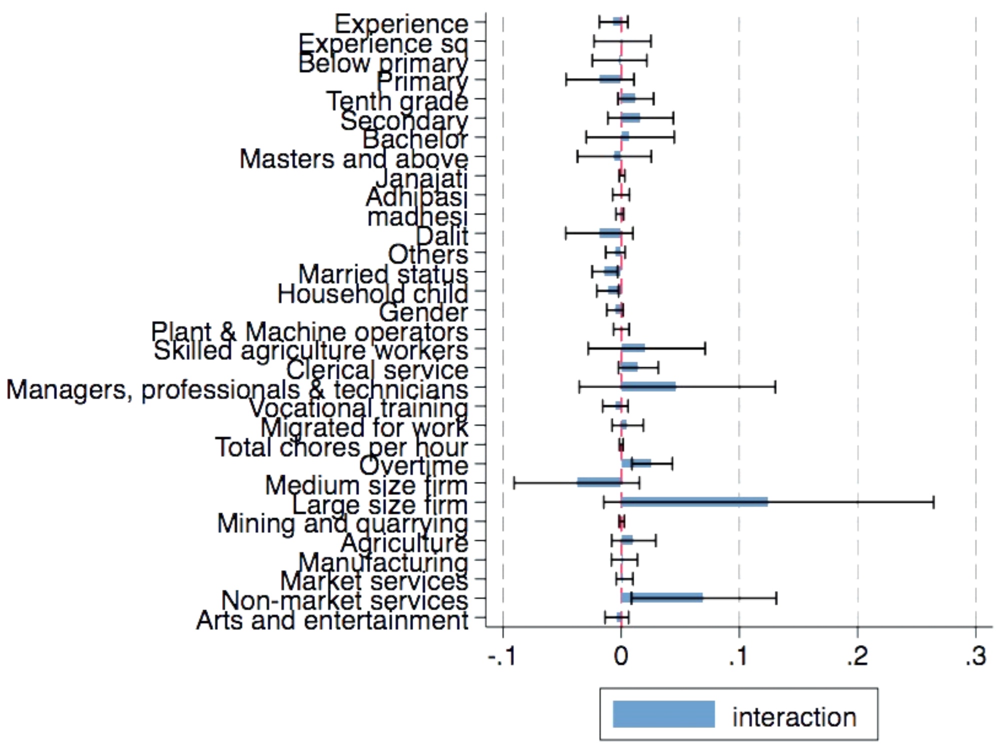


Figure 11 Threefold Blinder-Oaxaca Decomposition plot in mean wages for formal and informal employment across interaction



As the table 7 shows, the endowment component is 47.09 percent as part of the total difference in the wage gaps. Nevertheless, discrimination explains 17.29 percent lower wages and interaction explain about 35.62 percent lower wages for workers in informal employment than that of the workers in formal employment. Together the total attributable difference is close to 65 percent (64.38) between the workers in formal and informal employment.

Table 15 Earnings Gap Three Fold decomposition (Blinder-Oaxaca) between Formal employment and informal employment using different variables

	Endowments	%	Coefficients	%	Interaction	%	Total difference
experience	-0.019364*	-2.99	0.1141135	17.62	-0.0064933	-1.00	13.63
experience squared	0.0312548***	4.83	-0.0048964	-0.76	0.0010126	0.16	4.23
Education Level							
below primary	-0.0107466***	-1.66	0.0018233	0.28	-0.0013983	-0.22	-1.59
primary	-0.0219151***	-3.38	0.0279275	4.31	-0.0179599	-2.77	-1.84
tenth grade	0.0116325***	1.80	0.0243549*	3.76	0.0123095	1.90	7.46
secondary	0.0441931***	6.82	0.0131358	2.03	0.0163841	2.53	11.38
bachelor	0.1101066***	17.00	0.0032396	0.50	0.0076306	1.18	18.68
masters and above	0.1163745***	17.9	-0.0012559	-0.19	-0.0057880	-0.89	16.88
Caste group							
Janajati	-0.000326	-0.05	-0.0117549	-1.82	0.0006668	0.10	-1.76
Adhibasi	0.0062259***	0.96	0.0003321	0.05	-0.0001905	-0.03	0.98
Madhesi	0.0022452	0.35	0.0053855	0.83	-0.0012256	-0.19	0.99
Dalit	0.0052705	0.81	0.0210494	3.25	-0.0185196	-2.86	1.20
Others	0.0028267**	0.44	0.0069962	1.08	-0.0049631	-0.77	0.75
Married status	0.0068816***	1.06	-0.106069***	-16.38	-0.01385**	-2.14	-17.45
Household child	0.0017361	0.27	0.0359665***	5.55	-0.01148**	-1.77	4.05
Gender	0.0062975	0.97	0.0494156***	7.63	-0.0052953	-0.82	7.78
Occupation class							
Plant machine operator and assembler	-0.00527***	-0.81	-0.0001785	-0.03	0.0000773	0.01	-0.83
Skilled agriculture and trade workers	-0.0512687***	-7.92	-0.0239632	-3.70	0.0214997	3.32	-8.30
Clerical, service and sales workers	0.0012296	0.19	0.0175964*	2.72	0.0144748*	2.23	5.14
Managers, professional and technicians	0.100363***	15.50	0.0154916	2.39	0.0474817	7.33	25.22
Vocational training	0.0030921	0.48	-0.0054254	-0.84	-0.0049777	-0.77	-1.13
migration for work	0.0109443***	1.69	0.0051614	0.80	0.0055005	0.85	3.34
total chores per hour	-0.0000163	0.00	0.0157366	2.43	0.0000355	0.01	2.43
overtime	0.0320942***	4.96	-0.0710734***	-10.97	0.0261809***	4.04	-1.98

Size of business							
Medium size firm	-0.0097124**	-1.50	0.0438704	6.77	-0.0376485	-5.81	-0.54
Large size firm	0.0212552	3.28	0.0498085*	7.69	0.1247326*	19.26	30.23
Job Sector							
Mining and Quarrying; Electricity, gas and water supply	-0.002593	-0.40	0.0008240	0.13	0.0004275	0.07	-0.21
Agriculture	0.0316182***	4.88	-0.0174272	-2.69	0.010503	1.62	3.81
Manufacturing	0.0146576***	2.26	-0.0060233	-0.93	0.0026391	0.41	1.74
Market Services	-0.0098538*	-1.52	0.0143332	2.21	0.0028125	0.43	1.13
Non Market services	-0.1291066***	-19.93	0.0291996*	4.51	0.0699304*	10.80	-4.63
Arts, entertainment and other service activities	0.0048323***	0.75	0.0057689	0.89	-0.0037671	-0.58	1.06
constant	0.0000000	0.00	-0.1414946	-21.85	0.0000000	0.00	-21.85
Subtotal	**0.3049587**	47.09	**0.1119685**	17.29	**0.2307255**	35.62	**100.00**

Table 8 shows the relative contribution of each independent variables on the wage gap. The results shows the total wage gap attributed to difference in endowment, coefficient and interactions. Of the total wage gap, 47.09 percent is attributed to endowment, which is the mean increase in the hourly wage of employees in informal employment if they had the similar characteristics as the employees in formal employment. Similarly, 17.29 percent is attributed to the difference in coefficient, which quantifies the change in wages of employees in informal employment when applying the formal employment's coefficients to the informal employment's characteristics. Furthermore, 35.62 percent is attributed to interaction, which measures the simultaneous effect of differences in endowments and coefficients. This is also known as the unexplained part of the wage gap differences.

Table 9 provides the detailed decomposition of the endowment effects. The interpretation of the values is as: How would a worker in an informal employment earn more (>1) or less (<1) if the endowment effects were adjusted to the endowments of formal employment, evaluated at informal employment coefficients?

Experience is significant at 10 percent significance level and is associated with increase in hourly wage of 1 percent while squared of experience is significant at 1 percent significance level and on average earns 3 percent less than formal employment, adjusted for informal employment. Since age is already controlled for, experience is not biased with age in the labour market.

Education is significant at 1 percent significance level across all levels. With regards to distribution across education, where average wage level increases as education level increase, they show positive result for below primary and primary and negative result for rest of the educational level. An adjustment of informal employment levels to the one of the formal employments is associated with an economically valuable increase in hourly wages of 1-2 percent for below primary and primary education level and reduction in wages of 1–11 percent for rest of the education level.

With regard to those who are of Adhibasi caste group, and married, employees in informal employment would suffer minimal wage losses if they had the exactly same distribution across all variables as formal employment. There are only insignificant and negligible wage losses for the rest of caste group, having household child and gender of the employee.

The occupation class of the employees shows a particular interesting viewing. An adjustment of informal employment levels to the one of the formal employments shows minimal increase in hourly wages for plant machine operator and assemblers and 5 percent increase for skilled agriculture and trade workers. Meanwhile, the adjustment sees a huge decrease of 10 percent for managers, professional and technicians, while an insignificant and minimal decrease for clerical, service and sales workers. Vocational training and total household chores are insignificant, whereas migration for work and overtime shows the minimal wage loss of 1 and 3 percent respectively if employees in informal employment would have same distribution across those variables as employees in formal employment.

In terms of the size of firm, however, for medium size business, an adjustment of employees in informal employment levels to the one of formal employment is associated with a minimal valuable addition in wages for employees in informal employment, while the large size firm remains insignificant. Regarding to the job sector of employment, adjusting for informal employment levels to one of formal employment shows the 1 percent decrease for manufacturing, 3 percent decrease for agriculture and minimal decrease for arts, entertainment and other service activities. Similarly, the adjust shows a minimal increase for market services and 12.9 percent increment for non- market services. It means, if informal employment levels are adjusted to formal employment for non-market services, their earnings would be increase by Rs. 1669 per month. This shows the situation of market where most of the non-market activities are performed on a low-level payment.

In table 9, we present details of the coefficient effect. The interpretation of the table is as : How would an employee in formal employment earn more (>1) or less (<1) if the formal employment coefficients were adjusted to the coefficient level of employees in informal employment, evaluated at the mean formal employment endowment levels?

The coefficient effect shows the higher pay off emerging from the Nepalese labor market favoring formal employment over informal employment. It quantifies as the change in wages of employees of informal employment when applying the formal employment's coefficients to the informal employment's characteristics.

Among the education level, tenth grade is significant and the difference in payoffs for employees in informal employment is 2.4 percent less than formal employment leading to a wage gap of approximately Rs. 310 gross monthly income. The difference in rest of the education are insignificant and of negligible difference. One interesting thing is the minimal marginal gain for employees in informal employment over formal employment, if they have completed masters or above. Although, the difference in pay off for informal employment over formal employment decreases with the increase of educational level and eventually surpasses in masters and above, there are no noteworthy deviations in the evaluation by the employer based on educational level in the Nepalese labor market.

Similarly, for caste group in comparison to Khas, except for Janajati, the difference in pay off for informal employment over formal employment is minimal and insignificant. There seems to be highest of 5 percent discrimination in pay off for Dalit group, although it is insignificant. For the Janajati group, there is minimal and insignificant gain for employees in informal employment over formal employment. This implies there are no discrimination on the payoff for informal employment over formal employment in the labor market based on the caste group.

Married status is highly significant and the addition in payoff for married employees in informal employment over formal employment is 10.6 percent higher than unmarried employees leading to an approximate wage earning of Rs. 1372 per monthly income.

Household child is significant, with the payoff in wages for employees having children in informal employment over formal employment is 3.5 percent less than employees having no children employees leading to an approximate wage difference of Rs. 453 per monthly income. Gender of the employees is also highly significant, whereby female employees in informal employment over formal employment earn 4.9 percent less than male employees, leading to an approximate wage difference of Rs. 634 per month. There appears high and significant addition in payoffs for employees who are married and significant discrimination for female employees having household children. Females having to look after children and conduct household chores in the labor market suffers differences in payoff in informal employment over formal employment.

The occupation class is insignificant and the payoffs is minimal for informal employment over formal employment except clerical, service and sales workers. Keeping elementary occupation as the baseline category, employees as clerical, service and sales workers are significant, and the wage difference is minimal at 1.7 percent for informal employment over formal employment. Furthermore, there seems no discrimination from employers side based on having vocational trainings, having migrated for job and household chores performed. Having performed overtime is significant, and in comparison to employees not doing overtime job, one who does overtime in an informal employment over formal employment have an additional payoff of 7.1 percent leading to increase in wage of Rs. 919 per month.

Regarding the size of the business, keeping small size firm (< 5 employees) as the baseline category, medium size firm is insignificant, while large size firm is significant. There appears a wage gap difference of 4.98 percent for employees in informal employment over formal employment, resulting in the monthly wage difference of Rs. 645.

In terms of the nature of the job sector, only non-market services is significant, and there exists a wage gap difference of 2.9 percent for employees in informal employment over formal employment. It is interesting to note, among job sectors, only non-market services was the job sector with the gain for informal employment over formal employment, and the one with significant discrimination, resulting in wage difference of Rs. 375 per month. The rest of the job sector are insignificant and minimal. This suggests no difference in payoff based on coefficients among employees in formal and informal employment based on the job sector, except for non-market services.

Overall, the wage gap difference is highest (50.9 percent) with the differences in the educational attainment level, followed with large size firm (30.23 percent), managers, professional and technicians (25.22 percent) and experience (13.63 percent). For the educational attainment level, endowment effect as part of the total difference in wages is stronger at secondary and higher levels as compared to the endowment effect. Similarly, endowment effect is stronger for managers, professional and technicians too, compared to the discrimination effect. There exists higher discrimination effect based on the size of the business, with the higher the firm, higher is the discrimination between pay offs between formal and informal employment. The interaction between endowment effect and coefficient effect is highest for large size firm and non-market services. There is a scope for further study regarding the high unexplained component for large size firm and non-market services.

5. Conclusion and discussion

This study contributes to the current literature by defining informal sector employment and breaking down the disparities in earnings distribution between formal and informal sector employment for Nepal, where there is currently no empirical researches on informal sector employment and wages. This paper contributes to the study on whether there is differences in earning among employees in formal employment and informal employment in the Nepalese labor market. To this we have used the Nepal Labor Force Survey(NLFS) III, published by Central Bureau of Statistics (CBS) in 2018 covering approximately 77843 employees. We first carried out a series of (OLS) regressions to assess the role played by various variables in shaping hourly wages. Subsequently, we carried out a series of Oaxaca-Blinder decompositions to determine whether employees in informal employment faces a wage gap.

We have found that log hourly wage is positively significant to experience and educational level for formal employment. Similarly, log hourly wage is positively significant to experience, educational level and marital status while it is negatively significant to caste group except for Janajati, having household children and female informal employment. We have then included series of controls accounting for the job related characteristics in the basic model.

Among the occupation class, clerical, service and sales workers and managers, professionals and technicians are positively significant for formal employment on determining the log hourly wage rate. Similarly, all the occupation class is positively significant for wage rate, except for clerical, service and sales workers which is negatively significant on wage rate for informal employment. Workers migrated for work is positively significant, overtime work hour is negatively significant for formal and informal employment, whereas vocational training is positively significant for informal employment. Workers performing household chores is positively significant on formal employment and negatively significant on informal employment.

Including the control for the occupation class makes the Adhibasi caste group and marital status significant in the formal employment. Similarly, control for the occupation class makes Janajati caste group insignificant and control for migration for work makes household children insignificant in the informal employment. Including the control for the migration for work makes married status insignificance and increases the significance of household child in the formal employment. Moreover, the control for migration for work reduces the significance of vocational training in the informal employment. Moreover, the control for overtime makes marital status significant to 5 percent significance level, and household child less significant in the formal employment.

We have then included series of controls regarding the employer characteristics in the aforementioned model. The medium and large size of the firm is positively significant for formal employment, whereas only the medium firm is positively significant for the informal employment. All of the economic job sectors are positively significant except for arts, entertainment and other service activities in the formal employment and all the economic job sectors are positively significant except for non-market services in the informal employment. Including the control for the size of the firm, the significance of plant, machine operators and assemblers increases, significance of Dalit caste increases and decreases the significance of clerical, service and sales workers in the informal employment. Including the control for the job sector seems to make married status less significant and below primary education level insignificant in the formal employment. Furthermore, the control for job sectors increases the significance of large firm, reduces the significance of being Dalit.

The Blinder-Oaxaca decomposition results provide evidence of wage gap differences between employees in formal and informal employment. As the overall wage difference presents, employees in informal employment earn about 64.77 percent less than in formal employment do. Of the total wage gap, 47.09 percent is attributed to endowment, which is the mean increase in the hourly wage of employees in informal employment if they had the similar characteristics as the employees in formal employment. Similarly, 17.29 percent is attributed to the difference in coefficient, which quantifies the change in wages of employees in informal employment when applying the formal employment's coefficients to the informal employment's characteristics. Furthermore, 35.62 percent is attributed to interaction, which measures the simultaneous effect of differences in endowments and coefficients.

Overall, the wage gap difference is highest (50.9 percent) with the differences in the educational attainment level, followed with large size firm (30.23 percent), managers, professional and technicians (25.22 percent) and experience (13.63 percent). For the educational attainment level, endowment effect as part of the total difference in wages is stronger at secondary and higher levels. Similarly, endowment effect is stronger for managers, professional and technicians too, compared to the discrimination effect. There exists higher discrimination effect based on the size of the business, with higher the firm size, higher is the discrimination between pay offs between formal and informal employment. The interaction between endowment effect and coefficient effect is highest for large size firm and non-market services.

Our finding aligns with the previous literature where the level of education, number of years and experience causes the increase in the probability of working in the formal sector (Garcia et al., 2017), significantly higher in informal employment compared to the formal employment .Our study contradicts with the findings of (C. Williams & Gashi, 2022), on a study done in Kosovo, where the finding shows that wage gap for female is less than male, whereas our finding shows the higher wage gap for female in comparison to male in both the formal and informal employment.

The government of Nepal intends to formalize the informal sector workers through a focused social security and protection program . Increased pay for workers and increased productivity would result from policies aimed at transferring low-wage workers, agricultural laborers, and unskilled employees from informal to formal sectors where informality is dominant. Similarly, policy debates on educational reform have been ongoing for some time. Moving from primary to secondary and postsecondary education has significant policy implications for raising wages. Given that women generally earn less than males across all metrics, policies aimed at boosting female employees' assets are necessary. For Nepal to reduce the current pay gap, the shift to formality and protection is crucial.

If these people who are currently working informally and their employers are to be convinced to switch to formal employment, these results shows firstly, employers' costs in terms of fines and detection risks will need to rise dramatically. Second, it implies that the lack of formal work prospects for employees is what drives them to engage in informal employment, implying the need for proactive labor market policies that target vulnerable groups in order to enhance formal employment chances. Thirdly, if informality is to be reduced, since men make more money than women and majority of those who work in this sector, it may be necessary to target more men for these policies to detect and punish informal employment as well as active labor market policies to entice workers into the formal economy.

Although, government labor provisions are applicable to all working citizens, a prominent gap in policies and government interventions show exclusion of informal workers from existing mechanisms, hence wage inequality among these workers requires urgent policy attention. The increase in discrimination shows that despite increased liberalization and globalization as well as the state's withdrawal from several industries, discriminatory practices in informal employment have not decreased.. As a result, only a tiny portion of workers who are employed in formal employment can benefit from non-discriminatory laws leaving a large chunk of those employees in informal employment vulnerable, with lower protection and higher social, economic costs. Therefore, relying just on market forces to minimize the level of prejudice in the economy may not be sufficient. To stop the growing trend of wage discrimination, regulations must be put in place that cover those who work informally. There is an urgent need for social policies and policy reform to include informal workers as well the gradual formalization within the legal, social, economic and work benefits.

This analysis is limited to examining mean hourly wage differentials and its causes from a supply side. The findings indicate that along with formality or informality of employment, factors such as gender, educational level, experience, occupation class, size of business and job sector of employment affects the differences in payoffs. The combination of demand and supply side measures, investment to promote jobs in higher value added sectors, along with a focus on improved education, gender equality, and access to rights and benefits across job sector, occupation class and size of the business would gradually enhance the sectoral productivity as well as improved earning level of employees across informal employment.

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ANNEX

Research Matrix

Objectives	Hypothesis	Data Source	Outcome Variables	Data Analysis
1. To investigate the extent of wage gap difference between the formal and informal sectors in Nepal.	H_0 : There is no significant relationship between average wage rate and socio-economic and demographic factors. H_1: There is relationship between average wage and socio-economic and demographic factors	NLFS-III, 2017/18	Log Hourly Wage	Mincerian Wage function
2. To decompose wage gap difference of formal informal employment across gender	H_0: Difference in average wage rate of formal and informal employment is not explained by various socio-economic and demographic factors. H_1: Difference in average wage rate of formal and informal employment is explained by various factors.	NLFS-III, 2017/18	Log Hourly Wage	Oaxaca Decomposition

Table 16 Controls on the individual & household characteristics for formal employment

	Dependent variable: log(hourly wage)				
	1	2	3	4	5
experience	0.022***	0.023***	0.023***	0.022***	0.024***
	-0.003	-0.003	-0.003	-0.003	-0.003
experience squared	-0.0002***	-0.0002***	-0.0002***	-0.0002***	-0.0002***
	-0.0001	-0.0001	-0.0001	-0.0001	-0.0001
Education Level					
below primary	0.326***	0.317***	0.316***	0.315***	0.307***
	-0.079	-0.081	-0.081	-0.081	-0.081
primary	0.556***	0.539***	0.540***	0.537***	0.535***
	-0.071	-0.073	-0.073	-0.074	-0.074
tenth grade	0.912***	0.892***	0.894***	0.892***	0.891***
	-0.065	-0.068	-0.068	-0.068	-0.068
secondary	1.107***	1.085***	1.089***	1.087***	1.086***
	-0.066	-0.069	-0.069	-0.07	-0.07
bachelor	1.361***	1.337***	1.340***	1.338***	1.340***
	-0.067	-0.071	-0.072	-0.072	-0.072
masters and above	1.552***	1.527***	1.530***	1.528***	1.532***
	-0.07	-0.074	-0.074	-0.075	-0.075
Caste group					
Janajati		-0.026	-0.026	-0.025	-0.027
		-0.027	-0.027	-0.027	-0.027
Adhibasi		-0.08	-0.08	-0.079	-0.084
		-0.052	-0.052	-0.053	-0.053
Madhesi		-0.025	-0.03	-0.03	-0.031
		-0.039	-0.04	-0.04	-0.04
Dalit		0.057	0.055	0.054	0.057
		-0.081	-0.081	-0.082	-0.082
Others		-0.045	-0.047	-0.047	-0.049
		-0.103	-0.103	-0.104	-0.104
Household child			0.006	0.006	0.009
			-0.013	-0.014	-0.014
Gender				-0.009	-0.01
				-0.026	-0.026
Married status					-0.038
					-0.041
Constant	3.435***	3.469***	3.462***	3.468***	3.478***
	-0.068	-0.074	-0.076	-0.078	-0.079
Observations	1,153	1,153	1,153	1,153	1,153
R2	0.473	0.474	0.474	0.475	0.475
Adjusted R2	0.469	0.468	0.468	0.468	0.468
Residual Std. Error	8.153 (df = 1144)	8.157 (df = 1139)	8.159 (df = 1138)	8.163 (df = 1137)	8.163 (df = 1136)
F Statistic	128.113*** (df = 8; 1144)	79.070*** (df = 13; 1139)	73.384*** (df = 14; 1138)	68.445*** (df = 15; 1137)	64.212*** (df = 16; 1136)

Note: *p<0.1; **p<0.05; ***p<0.01

Table 17 Controls on the individual & household characteristics for informal employment

	Dependent variable: log(hourly wage)				
	1	2	3	4	5
experience	0.022***	0.022***	0.022***	0.021***	0.016***
	-0.001	-0.001	-0.001	-0.001	-0.001
experience squared	-0.0003***	-0.0003***	-0.0003***	-0.0003***	-0.0002***
	-0.00002	-0.00002	-0.00002	-0.00002	-0.00002
Education Level					
below primary	0.237***	0.220***	0.220***	0.143***	0.136***
	-0.017	-0.018	-0.018	-0.017	-0.017
primary	0.326***	0.307***	0.307***	0.198***	0.183***
	-0.017	-0.018	-0.018	-0.018	-0.018
teth grade	0.325***	0.298***	0.299***	0.219***	0.194***
	-0.021	-0.022	-0.022	-0.021	-0.021
secondary	0.491***	0.457***	0.457***	0.424***	0.392***
	-0.024	-0.025	-0.025	-0.024	-0.024
bachelor	0.857***	0.827***	0.827***	0.772***	0.736***
	-0.026	-0.028	-0.028	-0.027	-0.027
masters and above	1.139***	1.110***	1.110***	1.008***	0.970***
	-0.036	-0.037	-0.037	-0.036	-0.036
Caste group					
Janajati		0.042***	0.042***	0.028*	0.028**
		-0.015	-0.015	-0.014	-0.014
Adhibasi		-0.044**	-0.044**	-0.062***	-0.069***
		-0.019	-0.019	-0.018	-0.018
Madhesi		-0.042**	-0.042**	-0.094***	-0.095***
		-0.019	-0.019	-0.018	-0.018
Dalit		-0.037**	-0.037**	-0.053***	-0.060***
		-0.018	-0.018	-0.018	-0.018
Others		-0.041	-0.041	-0.100***	-0.101***
		-0.029	-0.029	-0.028	-0.028
Household child			0.0003	-0.005	-0.012**
			-0.005	-0.005	-0.005
Gender				-0.311***	-0.308***
				-0.012	-0.012
Married status					0.099***
					-0.014
Constant	3.627***	3.661***	3.660***	3.856***	3.862***
	-0.023	-0.027	-0.027	-0.027	-0.027
Observations	7,590	7,590	7,590	7,590	7,590
R2	0.186	0.19	0.19	0.26	0.265
Adjusted R2	0.185	0.189	0.189	0.258	0.263
Residual Std. Error	9.598 (df = 7581)	9.575 (df = 7576)	9.576 (df = 7575)	9.156 (df = 7574)	9.126 (df = 7573)
F Statistic	216.196*** (df = 8; 7581)	136.837*** (df = 13; 7576)	127.046*** (df = 14; 7575)	77.187*** (df = 15; 7574)	170.265*** (df = 16; 7573)
Note:	*p<0.1; **p<0.05; ***p<0.01				

Table 18 Controls on the job characteristics for formal employment

	Dependent variable: log(hourly wage)					
	1	2	3	4	5	6
experience	0.024***	0.022***	0.022***	0.022***	0.023***	0.022***
	-0.003	-0.003	-0.003	-0.003	-0.003	-0.003
experience squared	-0.0002***	-0.0002***	-0.0002***	-0.0002***	-0.0002***	-0.0002***
	-0.0001	-0.0001	-0.0001	-0.0001	-0.0001	-0.0001
Education Level						
below primary	0.307***	0.254***	0.255***	0.246***	0.247***	0.205***
	-0.081	-0.08	-0.08	-0.08	-0.079	-0.076
primary	0.535***	0.394***	0.398***	0.382***	0.387***	0.319***
	-0.074	-0.077	-0.077	-0.076	-0.076	-0.073
teth grade	0.891***	0.607***	0.613***	0.588***	0.592***	0.486***
	-0.068	-0.078	-0.078	-0.077	-0.077	-0.074
secondary	1.086***	0.754***	0.764***	0.729***	0.733***	0.617***
	-0.07	-0.081	-0.082	-0.081	-0.081	-0.078
bachelor	1.340***	0.965***	0.974***	0.936***	0.940***	0.853***
	-0.072	-0.085	-0.085	-0.085	-0.085	-0.081
masters and above	1.532***	1.152***	1.161***	1.121***	1.127***	1.010***
	-0.075	-0.087	-0.087	-0.087	-0.087	-0.084
Caste group						
Janajati	-0.027	-0.026	-0.027	-0.033	-0.034	-0.035
	-0.027	-0.026	-0.026	-0.026	-0.026	-0.025
Adhibasi	-0.084	-0.088*	-0.089*	-0.094*	-0.085*	-0.080*
	-0.053	-0.051	-0.051	-0.051	-0.051	-0.049
Madhesi	-0.031	-0.027	-0.025	-0.026	-0.022	-0.046
	-0.04	-0.039	-0.039	-0.039	-0.039	-0.037
Dalit	0.057	0.08	0.082	0.094	0.1	0.102
	-0.082	-0.079	-0.079	-0.079	-0.079	-0.075
Others	-0.049	-0.0003	-0.003	-0.007	-0.018	-0.022
	-0.104	-0.101	-0.101	-0.1	-0.1	-0.096
Household child	0.009	0.022	0.022	0.026*	0.025*	0.034***
	-0.014	-0.014	-0.014	-0.014	-0.014	-0.013
Gender	-0.01	-0.032	-0.031	-0.006	-0.035	-0.032
	-0.026	-0.026	-0.026	-0.026	-0.03	-0.029
Married status	-0.038	-0.069*	-0.070*	-0.048	-0.065	-0.082**
	-0.041	-0.04	-0.04	-0.04	-0.041	-0.039
Occupation class						
Plant machine operator and assembler		0.028	0.034	0.057	0.06	0.121*

		-0.071	-0.071	-0.071	-0.071	-0.068
Skilled agriculture and trade workers		0.029	0.034	0.018	0.022	0.028
		-0.075	-0.075	-0.075	-0.074	-0.071
Clerical, service and sales workers		0.204***	0.208***	0.203***	0.200***	0.197***
		-0.056	-0.056	-0.055	-0.055	-0.053
Managers, professional and technicians		0.383***	0.387***	0.397***	0.395***	0.362***
		-0.057	-0.057	-0.057	-0.057	-0.054
Vocational training			-0.032	-0.028	-0.034	-0.033
			-0.024	-0.024	-0.024	-0.023
migration for work				0.110***	0.104***	0.111***
				-0.025	-0.025	-0.024
total chores per hour					0.019**	0.008
					-0.01	-0.009
overtime						-0.235***
						-0.022
Constant	3.478***	3.551***	3.549***	3.507***	3.500***	3.756***
	-0.079	-0.08	-0.08	-0.08	-0.08	-0.08
Observations	1,153	1,153	1,153	1,153	1,153	1,153
R2	0.475	0.507	0.508	0.516	0.518	0.561
Adjusted R2	0.468	0.498	0.499	0.507	0.508	0.552
Residual Std. Error	8.163 (df = 1136)	7.924 (df = 1132)	7.921 (df = 1131)	7.857 (df = 1130)	7.846 (df = 1129)	7.487 (df = 1128)
F Statistic	64.212*** (df = 16; 1136)	58.203*** (df = 20; 1132)	55.551*** (df = 21; 1131)	54.792*** (df = 22; 1130)	52.731*** (df = 23; 1129)	60.168*** (df = 24; 1128)

Note: *p<0.1; **p<0.05; ***p<0.01

Table 19 Controls on the job characteristics for informal employment

	Dependent variable: log(hourly wage)					
	1	2	3	4	5	6
experience	0.016***	0.015***	0.015***	0.015***	0.015***	0.015***
	-0.001	-0.001	-0.001	-0.001	-0.001	-0.001
experience squared	-0.0002***	-0.0002***	-0.0002***	-0.0002***	-0.0002***	-0.0002***
	-0.00002	-0.00002	-0.00002	-0.00002	-0.00002	-0.00002
Education Level						
below primary	0.136***	0.112***	0.110***	0.112***	0.111***	0.113***
	-0.017	-0.017	-0.017	-0.017	-0.017	-0.017
primary	0.183***	0.155***	0.153***	0.154***	0.152***	0.153***
	-0.018	-0.017	-0.017	-0.017	-0.017	-0.017
teth grade	0.194***	0.177***	0.172***	0.172***	0.170***	0.171***
	-0.021	-0.022	-0.022	-0.022	-0.022	-0.022
secondary	0.392***	0.365***	0.358***	0.356***	0.354***	0.352***
	-0.024	-0.028	-0.029	-0.029	-0.029	-0.029
bachelor	0.736***	0.686***	0.683***	0.676***	0.675***	0.673***
	-0.027	-0.033	-0.033	-0.033	-0.033	-0.033
masters and above	0.970***	0.896***	0.891***	0.884***	0.883***	0.888***
	-0.036	-0.042	-0.042	-0.042	-0.042	-0.041
Caste group						
Janajati	0.028**	0.015	0.016	0.015	0.015	0.019
	-0.014	-0.014	-0.014	-0.014	-0.014	-0.014
Adhibasi	-0.069***	-0.077***	-0.077***	-0.074***	-0.076***	-0.071***
	-0.018	-0.018	-0.018	-0.018	-0.018	-0.018
Madhesi	-0.095***	-0.115***	-0.114***	-0.112***	-0.111***	-0.114***
	-0.018	-0.018	-0.018	-0.018	-0.018	-0.017
Dalit	-0.060***	-0.074***	-0.073***	-0.069***	-0.069***	-0.068***
	-0.018	-0.017	-0.017	-0.017	-0.017	-0.017
Others	-0.101***	-0.134***	-0.134***	-0.133***	-0.134***	-0.137***
	-0.028	-0.027	-0.027	-0.027	-0.027	-0.027
Household child	-0.012**	-0.008*	-0.008	-0.005	-0.003	-0.004
	-0.005	-0.005	-0.005	-0.005	-0.005	-0.005
Gender	-0.308***	-0.270***	-0.272***	-0.265***	-0.248***	-0.256***
	-0.012	-0.012	-0.012	-0.012	-0.014	-0.014
Married status	0.099***	0.077***	0.076***	0.080***	0.084***	0.082***
	-0.014	-0.014	-0.014	-0.014	-0.014	-0.014
Occupation class						
Plant machine operator and assembler		0.096***	0.090***	0.085***	0.083***	0.088***

		-0.021	-0.021	-0.021	-0.021	-0.021
Skilled agriculture and trade workers		0.242***	0.240***	0.238***	0.238***	0.242***
		-0.013	-0.013	-0.013	-0.013	-0.013
Clerical, service and sales workers		-0.055***	-0.056***	-0.060***	-0.061***	-0.061***
		-0.019	-0.019	-0.019	-0.019	-0.019
Managers, professional and technicians		0.154***	0.151***	0.150***	0.151***	0.124***
		-0.023	-0.024	-0.023	-0.023	-0.024
Vocational training			0.029**	0.027*	0.027*	0.021
			-0.015	-0.015	-0.015	-0.015
migration for work				0.063***	0.066***	0.072***
				-0.014	-0.014	-0.014
total chores per hour					-0.010**	-0.014***
					-0.004	-0.004
overtime						-0.089***
						-0.012
Constant	3.862***	3.812***	3.813***	3.794***	3.799***	3.869***
	-0.027	-0.027	-0.027	-0.027	-0.027	-0.028
Observations	7,590	7,590	7,590	7,590	7,590	7,590
R2	0.265	0.311	0.311	0.313	0.313	0.319
Adjusted R2	0.263	0.309	0.309	0.311	0.311	0.316
Residual Std. Error	9.126 (df = 7573)	8.838 (df = 7569)	8.836 (df = 7568)	8.826 (df = 7567)	8.823 (df = 7566)	8.789 (df = 7565)
F Statistic	170.265*** (df = 16; 7573)	170.556*** (df = 20; 7569)	162.690*** (df = 21; 7568)	156.533*** (df = 22; 7567)	150.041*** (df = 23; 7566)	147.402*** (df = 24; 7565)

Note: *p<0.1; **p<0.05; ***p<0.01

Table 20 Controls on the employer characteristics for formal employment

	Dependent variable: log(hourly wage)		
	1	2	3
experience	0.022***	0.022***	0.021***
	-0.003	-0.003	-0.003
experience squared	-0.0002***	-0.0002***	-0.0002***
	-0.0001	-0.0001	-0.0001
Education Level			
below primary	0.205***	0.202***	0.1
	-0.076	-0.075	-0.076
primary	0.319***	0.315***	0.220***
	-0.073	-0.072	-0.074
teth grade	0.486***	0.458***	0.340***
	-0.074	-0.074	-0.076
secondary	0.617***	0.585***	0.466***
	-0.078	-0.078	-0.08
bachelor	0.853***	0.821***	0.702***
	-0.081	-0.081	-0.083
masters and above	1.010***	0.975***	0.851***
	-0.084	-0.083	-0.085
Caste group			
Janajati	-0.035	-0.031	-0.03
	-0.025	-0.024	-0.024
Adhibasi	-0.080*	-0.090*	-0.080*
	-0.049	-0.048	-0.048
Madhesi	-0.046	-0.043	-0.035
	-0.037	-0.037	-0.037
Dalit	0.102	0.124*	0.091
	-0.075	-0.075	-0.074
Others	-0.022	0.055	0.075
	-0.096	-0.096	-0.095
Household child	0.034***	0.034***	0.035***
	-0.013	-0.013	-0.013
Gender	-0.032	-0.031	-0.032
	-0.029	-0.028	-0.028
Married status	-0.082**	-0.085**	-0.073*
	-0.039	-0.039	-0.038
Occupation class			
Plant machine operator and assembler	0.121*	0.183***	0.174**
	-0.068	-0.068	-0.068
Skilled agriculture and trade workers	0.028	0.009	0.042
	-0.071	-0.07	-0.07
Clerical, service and sales workers	0.197***	0.175***	0.169***
	-0.053	-0.052	-0.052
Managers, professional and technicians	0.362***	0.342***	0.320***
	-0.054	-0.054	-0.055
Vocational training	-0.033	-0.029	-0.025
	-0.023	-0.023	-0.023
migration for work	0.111***	0.111***	0.105***
	-0.024	-0.024	-0.023
total chores per hour	0.008	0.007	0.008
	-0.009	-0.009	-0.009
overtime	-0.235***	-0.232***	-0.228***

	-0.022	-0.022	-0.022
Size of business			
Medium size firm		0.293***	0.237***
		-0.08	-0.08
Large size firm		0.314***	0.215***
		-0.06	-0.063
Mining and Quarrying; Electricity, gas and water supply			0.264***
			-0.072
Construction			0.446***
			-0.091
Manufacturing			0.184***
			-0.058
Market Services			0.253***
			-0.053
Non Market services			0.281***
			-0.052
Arts, entertainment and other service activities			0.166
			-0.109
Constant	3.756***	3.501***	3.477***
	-0.08	-0.093	-0.092
Observations	1,153	1,153	1,153
R2	0.561	0.572	0.586
Adjusted R2	0.552	0.562	0.574
Residual Std. Error	7.487 (df = 1128)	7.404 (df = 1126)	7.302 (df = 1120)
F Statistic	60.168*** (df = 24; 1128)	57.828*** (df = 26; 1126)	49.504*** (df = 32; 1120)

Note: *p<0.1; **p<0.05; ***p<0.01

Table 21 Controls on the employer characteristics for informal employment

	Dependent variable: log(hourly wage)		
	1	2	3
experience	0.015***	0.015***	0.016***
	-0.001	-0.001	-0.001
experience squared	-0.0002***	-0.0002***	-0.0002***
	-0.00002	-0.00002	-0.00002
Education Level			
below primary	0.113***	0.112***	0.088***
	-0.017	-0.017	-0.016
primary	0.153***	0.152***	0.137***
	-0.017	-0.017	-0.017
teth grade	0.171***	0.171***	0.167***
	-0.022	-0.022	-0.021
secondary	0.352***	0.350***	0.345***
	-0.029	-0.029	-0.028
bachelor	0.673***	0.673***	0.667***
	-0.033	-0.033	-0.032
masters and above	0.888***	0.890***	0.904***
	-0.041	-0.041	-0.04
Caste group			
Janajati	0.019	0.02	0.02
	-0.014	-0.014	-0.013
Adhibasi	-0.071***	-0.074***	-0.087***
	-0.018	-0.018	-0.017
Madhesi	-0.114***	-0.115***	-0.075***
	-0.017	-0.017	-0.017
Dalit	-0.068***	-0.066***	-0.036**
	-0.017	-0.017	-0.017
Others	-0.137***	-0.136***	-0.098***
	-0.027	-0.027	-0.026
Household child	-0.004	-0.004	-0.006
	-0.005	-0.005	-0.005
Gender	-0.256***	-0.255***	-0.203***
	-0.014	-0.014	-0.014
Married status	0.082***	0.083***	0.070***
	-0.014	-0.014	-0.013
Occupation class			
Plant machine operator and assembler	0.088***	0.107***	0.167***
	-0.021	-0.021	-0.021
Skilled agriculture and trade workers	0.242***	0.245***	0.196***

	-0.013	-0.013	-0.013
Clerical, service and sales workers	-0.061***	-0.050**	0.013
	-0.019	-0.019	-0.021
Managers, professional and technicians	0.124***	0.138***	0.210***
	-0.024	-0.025	-0.027
Vocational training	0.021	0.021	0.023
	-0.015	-0.014	-0.014
migration for work	0.072***	0.064***	0.068***
	-0.014	-0.014	-0.014
total chores per hour	-0.014***	-0.013***	-0.007
	-0.004	-0.004	-0.004
overtime	-0.089***	-0.098***	-0.124***
	-0.012	-0.012	-0.011
bsizemedium_size_firm		0.087***	0.048***
		-0.012	-0.012
bsizelarge_size_firm		0.008	0.032*
		-0.016	-0.016
job_sectorMining_quarrying			0.127***
			-0.034
job_sectorConstruction			0.336***
			-0.017
job_sectorManufacturing			0.115***
			-0.02
job_sectorMarket_Services			0.080***
			-0.02
job_sectorNon_Market			0.006
			-0.026
job_sectorArts_ent			0.088***
			-0.031
Constant	3.869***	3.849***	3.703***
	-0.028	-0.028	-0.03
Observations	7,590	7,590	7,590
R2	0.319	0.323	0.368
Adjusted R2	0.316	0.321	0.365
Residual Std. Error	8.789 (df = 7565)	8.760 (df = 7563)	8.469 (df = 7557)
F Statistic	147.402*** (df = 24; 7565)	139.004*** (df = 26; 7563)	137.520*** (df = 32; 7557)

Note: *p<0.1; **p<0.05; ***p<0.01